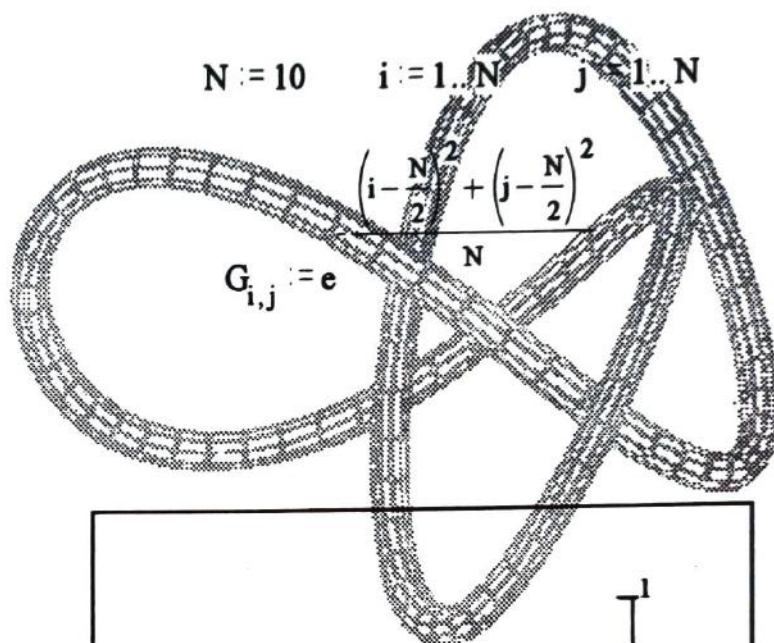
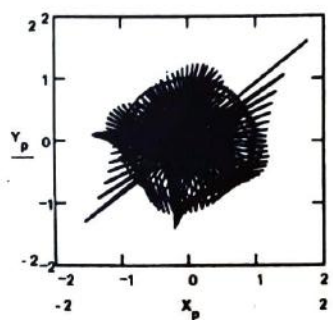
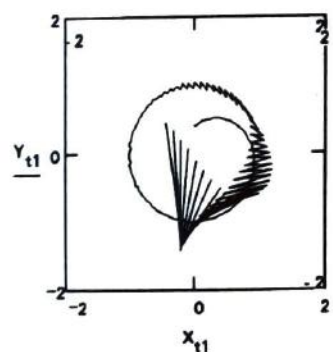
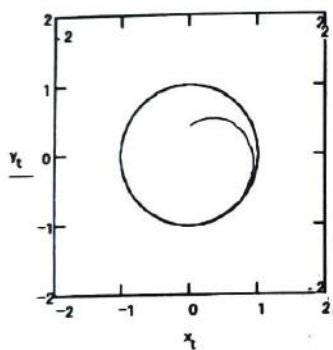
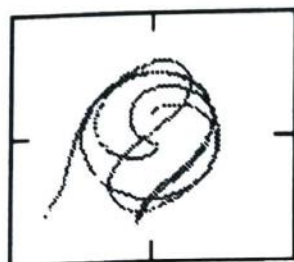


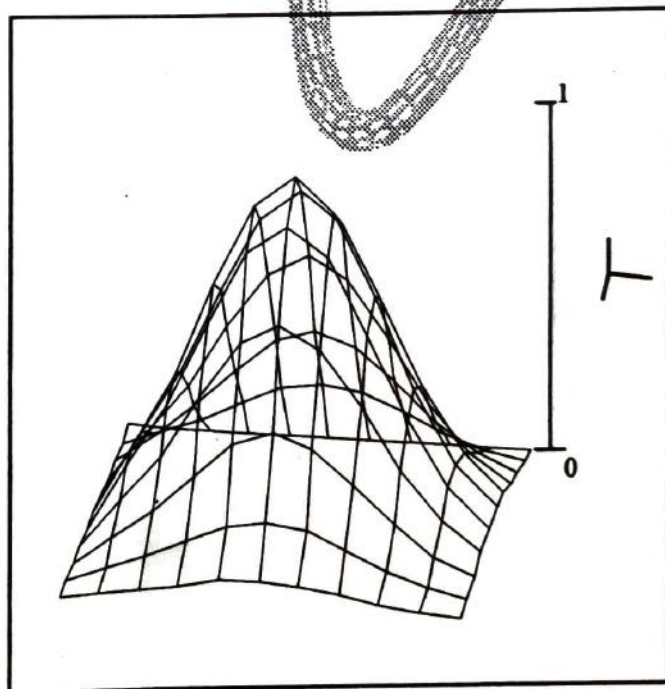
Math Survey



$$N = 10 \quad i = 1..N \quad j = 1..N$$

$$\left(i - \frac{N}{2}\right)^2 + \left(j - \frac{N}{2}\right)^2$$

$$G_{i,j} = e$$



G

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From the Editor-in-Chief

May 7, 1995

As editor-in-chief, I would like to thank you for reading Math Survey, Stuyvesant High School's official mathematics publication. My sincerest thanks go out to my dedicated staff members and editors who have contributed to the production of this publication. I would also like to take this opportunity to also thank Mr. Clancy, our adviser, and Dr. Richard Rothenberg, our "surrogate" adviser, for their advice and counsel in times of need. In my two years as editor-in-chief, they have been a great source of help and inspiration.

This school year, Math Survey has decided to publish twice a year instead of just annually. Our first issue was published mainly for the purpose of giving A.P. Calculus students a semester extra to study from our Calculus Review, while our **second** issue, "review-free," is for all to enjoy and ponder over. I hope all the math fans are immensely satisfied.

This is our first Spring issue of Math Survey, and I urge you to read and enjoy.

Sincerely,

A handwritten signature in cursive script that reads "Iris Lan".

Iris Lan

The Conjugacy Problem for Families of Matrices, With Applications to Fermat's Equation in Matrices and Theory of Symmetries on Affine Variables (Westinghouse version)

BY ALEXANDR KHAZANOV

I. Introduction

In this paper, we address simultaneous similarity of matrices by techniques based on a generalization of Hilbert's Theorem #90 and Elementary Lattice Theory. Let F be the matrix that induces a conjugation simultaneously transforming a family of matrices over a field extension K of $\mathbb{Q}$ into some family of matrices over $\mathbb{Q}$. We investigate conditions of existence of such matrix F by identifying rational matrices as fixed under transformations induced by the field automorphisms δ of K and then applying a generalization of Hilbert's Theorem #90 to the matrices $X_\delta = F(F^\delta)^{-1}$. The problem relating to the families of matrices over $\mathbb{Q}$ and to their simultaneous transformation into families of matrices over $\mathbb{Z}$ is addressed by associating the matrix F that induces a desired conjugation with the lattice generated by the column vectors of F . This approach reduces the problem of existence of conjugations to the problem of existence of invariant lattices.

The results derived concerning the conjugacy problem are then applied to Fermat's equation in matrices. Any solution of the Fermat's equation in matrices can be readily associated with a solution to the equation $A+B=C$ with certain restrictions on A , B , and C . Much of these restrictions is reflected in the characteristic polynomials of A , B , and C . Assume these polynomials to be fixed. If the characteristic polynomial of A is separable, then standard diagonalization theorems imply that the solution (A, B, C) can be obtained by conjugation from a solution (A_1, B_1, C_1) in matrices over a field extension K of $\mathbb{Q}$ where A_1 is diagonal and fixed and the characteristic polynomials of A_1 , B_1 , and C_1 are the same as the characteristic polynomials of A , B , and C respectively. Our results concerning the conjugacy problem are used to determine whether solutions (A_1, B_1, C_1) of the above form can be transformed by conjugation into solutions in matrices over $\mathbb{Q}$, and, ultimately, over $\mathbb{Z}$.

One of our results concerning Fermat's equation in matrices is that conjugacy classes of solutions to the system $x^3 = y^3 = (x+y)^3 = I$ in $SL_3\mathbb{Q}$ are in 1-1 correspondence with rational points on the affine variety determined by the equation $(2a^3 + 3a^2 + a + 1)^2 - 4w((3a^2 + 3a + 1)^2/3 - (3a^2 + 3a + 1)w + w^2) + 3r^2 = 0$. The transposition $(x, y) \mapsto (y, x)$ on the solutions to the above system of matrix equations, thus, induces a symmetry of period 2 on the set of rational points of this variety; this symmetry is not trivial and differs from the aw -plane symmetry. The afore mentioned symmetry can also be extended to an automorphism of period 2 on the whole variety over $\mathbb{C}$, and this automorphism commutes with the aw -plane symmetry induced by the transpose operation $(x, y) \mapsto (x^T, y^T)$.

Overall, a relationship between the conjugacy problem, equations in matrices and the symmetry problem for affine varieties is established, and some results concerning each of the three problems are obtained.

IA. Notations

- SL_3P — the group of all matrices of determinant 1 with elements in ring P ;
 GL_3P — the group of all invertible matrices with elements in ring P ;
 I — the identity matrix;
 x^δ — the value of field automorphism δ at x (where x is an element of the field);
 $\bar{x}$ — the value of the non-trivial field automorphism at x (where x is an element of a quadratic extension of $\mathbb{Q}$);
 F^δ — the matrix obtained by applying function δ to all elements of matrix F ;
 $p(G)$ — characteristic polynomial of the square matrix G ;
 Periodic matrix (resp. automorphism) — matrix (resp. automorphism) of finite order multiplicatively;
 Symmetry — periodic automorphism.

Throughout the paper the terms "rational matrix" and "integral matrix" will mean "matrix with rational elements" and "matrix with integer elements" respectively.

II. Statement of results

A) On the conjugacy question:

- (i) If $A_1, \dots, A_m \in SL_n\mathbb{Q}$ (respectively $GL_n\mathbb{Q}, M_n\mathbb{Q}$), then the following conditions are equivalent:
- 1⁰) There exists $F \in GL_n\mathbb{Q}$ such that $F^{-1}A_iF$ are integral matrices for all i ;
 - 2⁰) For any element C in the multiplicative semigroup G , generated by all $A_i (i = 1, \dots, m)$, $\text{tr}C \in \mathbb{Z}$;
 - 3⁰) There exists an effectively obtainable integer d such that $dC \in M_n\mathbb{Z}$ for any $C \in G$ (see 2⁰);
 - 4⁰) There exists an effective algorithm for finding $F \in GL_n\mathbb{Q}$ such that $F^{-1}A_iF \in SL_n\mathbb{Z}$ (respectively $GL_n\mathbb{Z}, M_n\mathbb{Z}$) for all i ;
- (ii) If $A, B \in SL_2\mathbb{Q}$, then the conditions of (i) are also equivalent to
- 5⁰) $\text{tr}A \in \mathbb{Z}, \text{tr}B \in \mathbb{Z}, \det(A+B) \in \mathbb{Z}$;
- (iii) Suppose $A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \lambda_i \in K = \mathbb{Q}(\sqrt{D}), \lambda_2 = \bar{\lambda}_1, B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then the following conditions are equivalent:
- 1⁰) There exists a matrix $F \in GL_2K$ such that $F^{-1}AF \in SL_2\mathbb{Q}, F^{-1}BF \in SL_2\mathbb{Q}$;
 - 2⁰) $\det A = \det B = 1; d = \bar{a}, N(a) - 1$ is the norm of an element in K ;
- (iv) For $\alpha \in \mathbb{Q}, m \in \mathbb{Q}$ the following conditions are equivalent:
- 1⁰) There exist A, B in $SL_2\mathbb{Q}$ such that $\text{tr}A = 2\alpha, \text{tr}B = 2m, A+B \in SL_2\mathbb{Q}$;
 - 2⁰) $(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1)$ is the norm of an element in $\mathbb{Q}(\sqrt{\alpha^2 - 1})$.
- (v) Let K is a cyclic extension of $\mathbb{Q}$ of degree n with generating automorphism δ and $A_1, \dots, A_r \in M_mK$. Then the following conditions are equivalent:
- 1⁰) There exists a matrix $F \in GL_mK$ such that $F^{-1}A_iF \in M_m\mathbb{Q}$ for all i ;

2°) There exists $X \in GL_m K$ such that $X^{-1}A_i X = A_i^\delta$ for all i and $XX^{\delta^1} \dots X^{\delta^{n-1}} = I$.

Conditions similar to the ones in (iii) and (iv) exist for higher dimensions as long as the Galois group of K over $\mathbb{Q}$ is cyclic (and even for arbitrary K — see section IV). However, they are much harder to deal with; the only result obtained through them is B)(ii).

B) On equation $x^n + y^n = z^n$ in $SL_2\mathbb{Z}$, $SL_3\mathbb{Z}$, $GL_3\mathbb{Z}$, and $SL_2\mathbb{Q}$:

- (i) In $SL_2\mathbb{Z}$ the equation has solutions if and only if n is not a multiple of 3 or 4;
- (ii) In $GL_3\mathbb{Z}$ or $SL_3\mathbb{Z}$ periodic solutions exist if and only if n is not a multiple of 3;
- (iii) For any solution $x^n + y^n = z^n$ in $SL_2\mathbb{Q}$ there exists a matrix f in $GL_2\mathbb{Q}$ such that for $z_1 = f^{-1}zf$, $y_1 = f^{-1}yf$, $z_1 = f^{-1}zf$, we have $x_1^n + y_1^n = z_1^n$ and $x_1^n \in SL_2\mathbb{Z}$, $y_1^n \in SL_2\mathbb{Z}$, $z_1^n \in SL_2\mathbb{Z}$, $z_1 \in SL_2\mathbb{Q}$, $y_1 \in SL_2\mathbb{Q}$, $z_1 \in SL_2\mathbb{Q}$;

(iv) For $n = 3$ solutions in $SL_2\mathbb{Q}$ exist.

C) On symmetries of a certain variety :

- (i) On the affine variety over $\mathbb{C}$ determined by the equation $(2a^3 + 3a^2 + a + 1)^2 - 4w((3a^2 + 3a + 1)w + w^3) + 3r^2 = 0$ there exists a non-trivial symmetry of period 2 that commutes with the aw -plane symmetry and transforms rational points into rational points.

III. Derivation of Results

Proofs of some technical lemmas are omitted due to lack of space. I would be honored to provide these proofs upon request. These proofs have been verified to be correct by reviewers of *Serdica*, professional math journal in which my paper based on the same research will be published in the first issue of 1995. The internet address of the journal is SERDICA%BGEARN@PSUVM.PSU.EDU. My paper will appear in it under the title "Fermat's Equation in Matrices".

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- 1. A Conjugacy Problem: Transformation of Families of Matrices Over a Field Extension of $\mathbb{Q}$ Into Families of Matrices Over $\mathbb{Q}$

Before we can proceed to our main results, we need the following

Lemma 1.1. (Generalization of Hilbert's Theorem #90 for matrices). Let K be a cyclic extension of $\mathbb{Q}$ of degree n with generating automorphism δ and $X \in GL_m K$. Then the following conditions are equivalent:

1°) $XX^{\delta^1} \dots X^{\delta^{n-1}} = I$;

2°) $\exists F \in GL_m K$ such that $X = F(F^\delta)^{-1}$

Proof. Despite the absence of the commutative property for matrix multiplication, the classical approach works here. 1° follows from 2° trivially, since

$$F(F^\delta)^{-1}(F(F^\delta)^{-1})^\delta \dots (F(F^\delta)^{-1})^{\delta^{n-1}} = F(F^\delta)^{-1}F^\delta(F^{\delta^2})^{-1} \dots F^{\delta^{n-1}}(F^{\delta^n})^{-1} = F(F^{\delta^n})^{-1} = FF^{-1} = I.$$

Suppose now that 1° holds. In our exponential notation $Y^\delta Y^\tau = Y^{\delta+\tau}$ ($\delta + \tau$ and $\tau + \delta$ may be different).

For any $\theta \in K$, we set

$$F(\theta) = \sum_{k=0}^{n-1} X^{1+\delta+\delta^2+\dots+\delta^{k-1}} \theta^{\delta^k}.$$

Note that for all $\theta \in K$,

$$\begin{aligned} X(F(\theta))^\delta &= X\left(\sum_{k=0}^{n-1} X^{1+\delta+\delta^2+\dots+\delta^{k-1}} \theta^{\delta^{k+1}}\right) = \sum_{k=0}^{n-2} X^{1+\delta+\delta^2+\dots+\delta^k} \theta^{\delta^{k+1}} + X^{1+\delta+\delta^2+\dots+\delta^{n-1}} \theta^{\delta^n} \\ &= \sum_{k=1}^{n-1} X^{1+\delta+\delta^2+\dots+\delta^{k-1}} \theta^{\delta^k} + I\theta^{\delta^n} = \sum_{k=0}^{n-1} X^{1+\delta+\delta^2+\dots+\delta^{k-1}} \theta^{\delta^k} = F(\theta). \end{aligned}$$

Our goal is to find $\theta \in K$ such that $F(\theta) \in GL_m K$, so that $X = F(\theta)((F(\theta))^\delta)^{-1}$. Let $d(\theta) = \det(F(\theta))$. We can express $d(\theta)$ as a linear combination of characters; the coefficient of $n \cdot id$ in this expression is equal to $\det I = 1$. By Artin's Theorem on the linear independence of characters, we obtain that $d(\theta) \neq 0$, hence $\exists \theta \in K$ such that $d(\theta) \neq 0$ and $\det F(\theta) \neq 0$. We let $F = F(\theta)$ to complete the proof.

Theorem 1.1. (*A(v) in the statement of results*). Let K be a cyclic extension of $\mathbb{Q}$ of degree n with generating automorphism δ and $A_1, \dots, A_r \in M_m K$. Then the following conditions are equivalent:

1°) There exists $F \in GL_m K$ such that $F^{-1}A_i F \in M_m \mathbb{Q}$ for all i .

2°) There exists $X \in GL_m K$ such that $X^{-1}A_i X = A_i^\delta$ for all i and $XX^{\delta^1} \dots X^{\delta^{n-1}} = I$.

Proof. By Lemma 1.1, condition 2° is equivalent to the existence of $F \in GL_m K$ such that

$$(F(F^\delta)^{-1})^{-1}A_i F(F^\delta)^{-1} = A_i^\delta$$

for all i , or, equivalently, $F^{-1}A_i F = (F^\delta)^{-1}A_i^\delta F^\delta$ for all i , or, equivalently, 1°, since δ generates the Galois group of K over $\mathbb{Q}$.

a. The case of 2 by 2 matrices

Theorem 1.2. (*Theorem IIA)(iii) in the statement of results*.) Suppose $A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, $\lambda_i \in K = \mathbb{Q}(\sqrt{D})$, $\lambda_2 = \bar{\lambda}_1$, $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then the following conditions are equivalent:

1°) There exists a matrix $F \in GL_2 K$ such that $F^{-1}AF \in SL_2 \mathbb{Q}$, $F^{-1}BF \in SL_2 \mathbb{Q}$;

2°) $\det A = \det B = 1$; $d = \bar{a}$, $N(a) - 1$ is the norm of an element in K .

Proof. We rewrite 1° according to Theorem 1.1. We note that $A^\delta = \bar{A} = \begin{pmatrix} \lambda_2 & 0 \\ 0 & \lambda_1 \end{pmatrix}$, i.e., $X^{-1}AX = \bar{A}$ can be rewritten as $\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} X = X \begin{pmatrix} \lambda_2 & 0 \\ 0 & \lambda_1 \end{pmatrix}$. Thus, if $X = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ then $e = h = 0$. Since $X\bar{X} = \begin{pmatrix} 0 & f \\ g & 0 \end{pmatrix} \begin{pmatrix} 0 & \bar{f} \\ \bar{g} & 0 \end{pmatrix} = \begin{pmatrix} f\bar{g} & 0 \\ 0 & \bar{f}g \end{pmatrix}$, $X\bar{X} = I$ is equivalent to $f\bar{g} = 1$. Finally, $X^{-1}BX = \bar{B}$ can be rewritten as $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & f \\ g & 0 \end{pmatrix} = \begin{pmatrix} 0 & f \\ g & 0 \end{pmatrix} \begin{pmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{pmatrix}$, i.e., as system of equations:

$$(1) bg = f\bar{c};$$

$$(2) af = f\bar{d};$$

$$(3) dg = g\bar{a};$$

$$(4) cf = g\bar{b}.$$

Since $f\bar{g} = 1$, (2) and (3) are both equivalent to $d = \bar{a}$. (1) and (4) can be rewritten as $N(g) = g\bar{g} = gf^{-1} = c(\bar{b})^{-1} = \bar{c}b^{-1} = N(c)(bc)^{-1}$. Thus, 1° implies $d = \bar{a}$ and $N(a) - 1 = ad - 1 = bc = N(cg^{-1})$, that is 1° implies 2° . Conversely, 2° implies (2) and (3) and the existence of $s \in K$ such that $N(s) = N(a) - 1$. We let $g = cs^{-1}$, $f = (\bar{g})^{-1}$ to obtain matrix $X = \begin{pmatrix} 0 & f \\ g & 0 \end{pmatrix}$ satisfying the necessary conditions. Condition $\det A = \det B = 1$ assures that the resulting conjugates in M_2Q in fact lie in SL_2Q .

1A. Applications to Fermat's equation in matrices

a. The case of 2 by 2 matrices

Theorem 1A.1. (Theorem IIA)(iv) in the statement of results). For $\alpha \in Q$, $m \in Q$, the following conditions are equivalent:

1°) There exist A, B in SL_2Q such that $\text{tr}A = 2\alpha$, $\text{tr}B = 2m$, $A + B \in SL_2Q$;

2°) $(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1)$ is the norm of an element in $Q(\sqrt{\alpha^2 - 1})$.

Proof. We will need the following lemma, whose proof we omit due to lack of space:

Lemma 1A.1. Suppose $A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, $\lambda_i \in K = Q(\sqrt{D})$, $\lambda_2 = \bar{\lambda}_1$, $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$; $\det A = \det B = 1$, $\text{tr}A = 2\alpha \in Q$, $\text{tr}B = 2m \in Q$; $D = \alpha^2 - 1$; $a = m_1 + n\sqrt{D}$, $m_1, n \in Q$; $\lambda_1 = \alpha + \sqrt{D}$. Then the following conditions 1° and 2° are equivalent if $\sqrt{D} \notin Q$ (1° follows from 2° in any case):

$$1^\circ) \det(A + B) = 1;$$

$$2^\circ) m_1 = m, n = (\alpha m + 1/2)/D.$$

If $K = Q(\sqrt{D}) \neq Q$, then 2° can also be rewritten as $d = \bar{a}$, $n = (\alpha m + 1/2)/D$.

Proof of Theorem 1A.1. If $\alpha = \pm 1$, then both conditions hold for any m . (Indeed, if $\alpha m \neq -1/2$, we can set $A = \begin{pmatrix} \alpha & \alpha \\ 0 & \alpha \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1/(2m + \alpha) \\ 2m + \alpha & 2m \end{pmatrix}$; if $\alpha m = -1/2$, we can set $A = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$, $B = \begin{pmatrix} 0 & -\alpha \\ \alpha & -\alpha \end{pmatrix}$, thus showing 1° ; 2° holds since $(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1) = (\alpha m + 1/2)^2$.)

$$1^\circ \Rightarrow 2^\circ. \text{ If } \alpha \neq \pm 1, A \text{ is conjugate to } A_1 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \text{ where } \lambda_1 = \alpha + \sqrt{D}, \lambda_2 = \alpha - \sqrt{D}, D = \alpha^2 - 1.$$

The conjugate B_1 to B under the same conjugation can be written as $B_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d \in Q(\sqrt{D})$, i.e., $a = m_1 + n\sqrt{D}$ for some rational m_1, n . We have $\det A_1 = \det A = 1$, $\det B_1 = \det B = 1$, and $\det(A_1 + B_1) = \det(A + B) = 1$. Now A_1 and B_1 satisfy condition 1° of Theorem 1.2 and condition 1° of Lemma 1A.1. If $\sqrt{D} \in Q$, condition 2° is obviously satisfied. If $\sqrt{D} \notin Q$, by Lemma 1A.1, we obtain $m = m_1$. Also, $n = (\alpha m + 1/2)/D$. Thus,

$$N(a) - 1 = m^2 - n^2D - 1 = m^2 - 1 - \frac{(\alpha m + 1/2)^2}{(\alpha^2 - 1)^2} \cdot (\alpha^2 - 1)$$

$$= \left(\frac{1}{1-\alpha^2} \right) \cdot ((\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1)).$$

By Theorem 1.2, $N(a) - 1$ is the norm of an element in $K = \mathbb{Q}(\sqrt{\alpha^2 - 1})$. Also,

$$(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1) = (1 - \alpha^2)(N(a) - 1) = N(\sqrt{\alpha^2 - 1})(N(a) - 1),$$

from which 2° follows.

2° $\Rightarrow$ 1°. Conversely, we can consider $A_1 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, $\lambda_1 = \alpha + \sqrt{D}$, $\lambda_2 = \alpha - \sqrt{D}$, $D = \alpha^2 - 1$; $B_1 = \begin{pmatrix} m + n\sqrt{D} & b \\ 1 & m - n\sqrt{D} \end{pmatrix}$, where $n = (\alpha m + 1/2)/D$, $b = m^2 - 1 - n^2 D$. A_1 and B_1 satisfy conditions 2° of Lemma 1A.1 and 2° of Theorem 1.2. Thus by Theorem 1.2, $\exists F \in GL_2$ of $\mathbb{Q}(\sqrt{D})$ such that $F^{-1}A_1F \in SL_2\mathbb{Q}$ and $F^{-1}B_1F \in SL_2\mathbb{Q}$. By Lemma 2.1, $A_1 + B_1 \in SL_2\mathbb{Q}(\sqrt{D})$; hence, $F^{-1}A_1F + F^{-1}B_1F = F^{-1}(A_1 + B_1)F \in SL_2\mathbb{Q}$. We let $A = F^{-1}A_1F$, $B = F^{-1}B_1F$ to complete the proof.

Corollary 1. Suppose, $\alpha, m \in \mathbb{Q}$; $2\alpha, 2m \in \mathbb{Z}$; $2\alpha \equiv 2m \equiv 0 \pmod{9}$. Then there do not exist $A, B \in SL_2\mathbb{Q}$ such that $\text{tr}A = 2\alpha$, $\text{tr}B = 2m$, and $A + B \in SL_2\mathbb{Q}$.

Proof. Assume otherwise. By Theorem 1A.1, that would imply

$$(1) (\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1) = x^2 - (\alpha^2 - 1)y^2 \text{ with } x, y \in \mathbb{Q}.$$

Let $d = \text{l.c.d.}(x, y)$. (1) can be rewritten as

$$(2) d^2((2\alpha \cdot 2m + 2)^2 - ((2\alpha)^2 - 4)((2m)^2 - 4)) = (4xd)^2 - ((2\alpha)^2 - 4)(2yd)^2, \quad xd, yd \in \mathbb{Z}.$$

Now, $(2\alpha \cdot 2m + 2)^2 - ((2\alpha)^2 - 4)((2m)^2 - 4) \equiv 2^2 - (-4)(-4) \equiv -12 \equiv 6 \pmod{9}$. Therefore,

$$(4xd)^2 + (2yd)^2 \equiv (4xd)^2 - ((2\alpha)^2 - 4)(2yd)^2 \equiv d^2((2\alpha \cdot 2m + 2)^2 - ((2\alpha)^2 - 4)((2m)^2 - 4)) \equiv 0 \pmod{3}.$$

That implies $4xd \equiv 2yd \equiv 0 \pmod{3}$; hence, the right side in (2) is 0 modulo 9. Since the factor besides d^2 in the left side of (2) is congruent to 6 modulo 9, we conclude that $d \equiv 0 \pmod{3}$. Together with $xd \equiv yd \equiv 0 \pmod{3}$, that contradicts $d = \text{l.c.d.}(x, y)$ and completes the proof.

Corollary 2. The equation $x^n + y^n = z^n$ has no solutions in $SL_2\mathbb{Z}$ if n is a multiple of 3.

Proof. It is sufficient to prove it for $n = 3$. By Corollary 1, everything follows from the following

Lemma 1A.2. If $A^3 + B^3 = C^3$ with $A, B, C \in SL_2(\mathbb{Z}/9\mathbb{Z})$ then $\text{tr}A^3 = \text{tr}B^3 = \text{tr}C^3 = 0$.

Proof. Let $a = \text{tr}A$, $b = \text{tr}B$, $c = \text{tr}C$. Then $\text{tr}A^3 = a^3 - 3a$, $\text{tr}B^3 = b^3 - 3b$, $\text{tr}C^3 = c^3 - 3c$. In $\mathbb{Z}/9\mathbb{Z}$, $x^3 - 3x$ is 0, -2, and 2 for $x \equiv 0, 1, 2 \pmod{3}$ respectively. Since $\text{tr}A^3 + \text{tr}B^3 = \text{tr}C^3$, we obtain that either $\text{tr}A^3$, $\text{tr}B^3$, or $\text{tr}C^3$ equals 0. Unless $\text{tr}A^3 = \text{tr}B^3 = \text{tr}C^3 = 0$, the equality can be rearranged and values of variables can be redefined so that $\text{tr}A^3 = 0$, $\text{tr}B^3 = 2$, $\text{tr}C^3 = 2$. We must prove that the latter is impossible. We now consider matrices modulo 3. The characteristic polynomials of B and C modulo 3 are $(t - 1)^2$. Hence, B and C are conjugate to matrices of the form $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$, whose cube is I . B^3 and C^3 are thus conjugate to I , i.e., $B^3 = C^3 = I$; hence, $A^3 = 0$, i.e., $(\det A)^3 = \det A^3 = 0$; hence, $\det A = 0 \pmod{3}$. This is a contradiction. Therefore, $\text{tr}A^3 = \text{tr}B^3 = \text{tr}C^3 = 0$, q.e.d.

Remark 1. Our proof of Corollary 2 with slight modifications actually proves that solutions do not exist in matrices over the field of 3-adic numbers. This means that solutions do not exist over the residue ring

modulo some power of 3. It may actually be shown (by direct inspection if not in any better way) that solutions do not exist modulo 9. However, our method is more general than such inspection and leads to a number of results other than the above corollary.

Corollary 3. (Theorem IIB)(i) in the statement of results) the equation $x^n + y^n = z^n$ has solutions in $SL_2\mathbb{Z}$ if and only if n is not a multiple of either 3 or 4.

Proof. The "only if" part follows directly from Corollary 2 and the fact that $x^4 + y^4 = z^4$ has no solutions in $SL_2\mathbb{Z}$. The proof of the latter fact, based on the consideration of traces modulo 8 can be found in [1]. The solutions for the "if" part can also be found (indirectly) in [1]. However, we will demonstrate how our general approach based on the proof of Lemma 1.1, Theorem 1.1, Theorem 1.2, and Theorem 1A.1 and its Corollary 1 works to produce the solutions. We shall also derive that equation $x^3 + y^3 = z^3$ has solutions in $SL_2\mathbb{Q}$ (result IIB)(iv)).

To find solutions in $SL_2\mathbb{Q}$, we need to find α and m satisfying the conditions of Theorem 1A.1 such that α , m , and $\alpha + m$ are not equal to 1 and $A, B, C \in SL_2\mathbb{Q}$ such that $\text{tr}A^n = 2\alpha$, $\text{tr}B^n = 2m$, and $\text{tr}C^n = 2(\alpha + m)$. We can then apply an appropriate conjugation to A, B , and C to obtain the solutions of the equation. For $z \in SL_2\mathbb{Q}$, $\text{tr}z^3 = (\text{tr}z)^3 - 3\text{tr}z$. If $\text{tr}z = 0$, then $\text{tr}z^3 = 0$. If we could find $a \in \mathbb{Q}$ such that $\alpha = 0$ and $m = (a^3 - 3a)/2$ satisfy the conditions of Theorem 1A.1, we would prove the existence of solutions of $x^3 + y^3 = z^3$ in $SL_2\mathbb{Q}$. For $\alpha = 0$, $(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1) = m^2 - 3/4$. For $a = 4$, $m = 26$, $(\alpha m + 1/2)^2 - (\alpha^2 - 1)(m^2 - 1) = 26^2 - 3/4 = 676 - 3/4 = 675 3/4 = 51^2/4 + 5\sqrt{-1}$. Thus, solutions of the equation exist. One such solution, obtained by applying our method, is

$$x = \begin{pmatrix} -1 & -2 \\ 1 & 1 \end{pmatrix}, y = \begin{pmatrix} \frac{3}{26} & \frac{3}{13} \\ \frac{17}{26} & \frac{17}{13} \end{pmatrix}, z^3 = \begin{pmatrix} 6 & 11 \\ 25 & 46 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} + \begin{pmatrix} 5 & 9 \\ 26 & 47 \end{pmatrix} = x^3 + y^3$$

(z can be obtained from y by an appropriate conjugation).

To find periodic solutions of $x^n + y^n = z^n$ for $n \equiv \pm 2 \pmod{12}$, we set $\alpha(z^2) = -1/2$, $m(y^2) = -1/2$ (these values, as well as similar values below, were obtained by analyzing characteristic polynomials of periodic matrices through eigenvalues). Following the procedure of Theorem 1A.1, Lemma 1.1, etc., we obtain:
 $x = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$, $y = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$, $z = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$ ($n \equiv 2 \pmod{12}$). Take their inverse for $n \equiv -2 \pmod{12}$.

To find periodic solutions for odd n not divisible by 3, we use $\alpha(z) = -1/2$, $m(y) = -1/2$ and obtain the solution: $x = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$, $y = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$, $z = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ ($n \equiv 1 \pmod{6}$). Take their inverse for $n \equiv 5 \pmod{6}$.

b. The case of 3 by 3 matrices

Theorem 1A.2. Let $\rho = (-1 + \sqrt{-3})/2 = \sqrt[3]{1}$ and $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in M_3\mathbb{Q}(\rho)$. Let $\Lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & 0 & \rho^2 \end{pmatrix}$, $u = bd$, $v = cg$, $w = fh$, $p = bfg$, $q = cdh$. Then the following conditions 1° and 2° are equivalent:

1°)

$$a) p(A) = p(A + \Lambda) = t^3 - 1;$$

b) $\exists F \in GL_3\mathbb{Q}(\rho)$ such that $F^{-1}AF \in SL_3\mathbb{Q}$, $F^{-1}\Lambda F \in SL_3\mathbb{Q}$;

2°)

(a) $e = a\rho$, $i = a\rho^2$,

(b) $u + v + w = 0$,

(c) $u\rho^2 + v\rho + w = 3a^2 + 3a + 1$,

(d) $p + q = 2a^3 + 3a^2 + a + 1$.

(e) $a, w \in \mathbb{Q}$; $q = \bar{p}$.

Proof. It may be proven that the combination of parts ((a), (b), (c), (d)) of condition 2° is equivalent to part (a) of condition 1°. (We omit the proof – see Lemma 2.2 in [8]). We must prove that this combination (or equivalently part (a) of 1°) provided, part (e) of condition 2° should be equivalent to part (b) of condition 1°. We use Theorem 1.1. Since $\bar{\Lambda} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & \rho \end{pmatrix}$, condition $X^{-1}\Lambda X = \bar{\Lambda}$ can be rewritten

as $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & 0 & \rho^2 \end{pmatrix} X = X \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & \rho \end{pmatrix}$ to imply that X is of the form $\begin{pmatrix} \alpha & 0 & 0 \\ 0 & 0 & \beta \\ 0 & \gamma & 0 \end{pmatrix}$. Condition $X\bar{X} = I$ can be rewritten as $N(\alpha) = 1$, $\beta\bar{\gamma} = 1$. Condition $X^{-1}\Lambda X = \bar{\Lambda}$ can be rewritten as $AX = X\bar{A}$, i.e., $\begin{pmatrix} a\alpha & c\gamma & b\beta \\ d\alpha & f\gamma & e\beta \\ g\alpha & i\gamma & h\beta \end{pmatrix} = \begin{pmatrix} \alpha\bar{a} & \alpha\bar{b} & \alpha\bar{c} \\ \beta\bar{g} & \beta\bar{h} & \beta\bar{i} \\ \gamma\bar{d} & \gamma\bar{e} & \gamma\bar{f} \end{pmatrix}$ which is equivalent to the following system of equations:

(1) $a\alpha = \alpha\bar{a}$;

(2) $c\gamma = \alpha\bar{b}$;

(3) $b\beta = \alpha\bar{c}$;

(4) $\beta\bar{g} = d\alpha$;

(5) $\beta\bar{h} = f\gamma$;

(6) $\beta\bar{i} = e\beta$;

(7) $\gamma\bar{d} = g\alpha$;

(8) $\gamma\bar{e} = i\gamma$;

(9) $h\beta = \gamma\bar{f}$.

We now derive 2° from 1°. Assuming 1°, we also prove that $u \neq 0$, $v \neq 0$, $w \neq 0$.

$1^\circ \Rightarrow 2^\circ$. (1) implies $a = \bar{a}$, i.e., $a \in \mathbb{Q}$, since $\alpha \neq 0$ as $N(\alpha) = 1$. (4) implies $b\beta\bar{g} = b\bar{d}\alpha$, and (3) implies $b\beta\bar{g} = \alpha\bar{c}\bar{g}$. Thus, together (3) and (4) imply $b\bar{d}\alpha = b\beta\bar{g} = \alpha\bar{c}\bar{g}$; since $\alpha \neq 0$, that implies: $b\bar{d} = \bar{c}\bar{g}$, i.e., $u = \bar{v}$. By (2°)(b), we have: $w = -(u + v) = -(\bar{v} + v) \in \mathbb{Q}$. If $w = 0$, then either $f = 0$ or $h = 0$. Hence, by (9), since $\beta \neq 0$, $\gamma \neq 0$ (as $\beta\bar{\gamma} = 1$), we would obtain that $f = h = 0$. That would mean $p = q = 0$; by 2°(d), that would imply $2a^3 + 3a^2 + a + 1 = 0$, which is impossible for $a \in \mathbb{Q}$. Thus, $w \neq 0$. Since $u = \bar{v}$ and $u + v = -w$, we also obtain that $u \neq 0$, $v \neq 0$. It follows from (5) that $N(\beta h) = \beta\bar{h}\bar{\beta}h = f\gamma\bar{\beta}h = fh = w = pq/uv = pq/N(u)$. Since $u \neq 0$, condition (3) implies that $\beta = \alpha\bar{c}b^{-1}$.

Hence, $N(\beta h) = N(\alpha\bar{c}h/b) = N(ch/b) = N(q/u) = N(q)/N(u)$. Since $pq = uvw \neq 0$, we have $q \neq 0$. Thus, from $pq = N(u)(pq/N(u)) = N(u)N(\beta h) = N(u)(N(q)/N(u)) = N(q) = q/\text{bar } q$, it

follows that $p = \bar{q}$. Thus, $2^\circ(e)$ follows from 1° which completes the proof.

Conversely, 2° implies 1° . To prove this, we are to prove that under (2°) , $\exists \alpha, \beta, \gamma$ such that $N(\alpha) = 1$, $\beta\bar{\gamma} = 1$, and conditions (1)–(9) hold.

The condition that $w, a \in Q$, in conjunction with conditions (b) and (c), implies that $v = \bar{u}$. Also, if either $u = 0$, $v = 0$, or $w = 0$, then either $p = 0$ or $q = 0$. Thus, since $q = \bar{p}$, we would obtain $p = q = 0$, which contradicts condition (d). Thus, $u \neq 0$, $v \neq 0$, $w \neq 0$; hence, $b, d, c, g, f, h, p, q \neq 0$. We now set arbitrarily $\alpha \in Q(\rho)$ so that $N(\alpha) = 1$; β can then be set according to (3): $\beta = (\alpha\bar{c}/b) \neq 0$. We set $\gamma = 1/\bar{\beta}$. Now, we must prove that conditions (1)–(9) are satisfied.

(1) holds since $a = \bar{a}$ as $a \in Q$;

(2) holds as $c\gamma = c/\bar{\beta} = c\bar{b}/(\bar{\alpha}c) = \alpha\bar{b}$;

(3) holds by definition of β ;

(4) holds since $\beta\bar{g} = \alpha\bar{c}g/b = \alpha\bar{u}/b = \alpha u/b = \alpha d$;

(5) holds since $\beta\bar{h} = N(\beta h)/(\bar{\beta}h) = \gamma N(\beta h)/h = f\gamma N(\beta h)/w = (f\gamma/w)N(h\alpha\bar{c}/b) = (f\gamma/w)N(hc/b) = (f\gamma/w)N(q/u) = (f\gamma/w)q\bar{q}/(u\bar{u}) = (f\gamma/w)qp/(uv) = (f\gamma/w)w = f\gamma$;

(6) holds since $\beta\bar{i} = \beta e$, as $a \in Q$, $e = a\rho$, $i = a\rho^2$;

(7) holds since $\gamma\bar{d} = \bar{b}\alpha\bar{d}/c = \alpha\bar{b}\bar{d}/c = \alpha\bar{u}/c = \alpha v/c = \alpha g$;

(8) holds since $\gamma\bar{e} = \gamma i$, since $\bar{e} = i$ (see (6))

(9) holds since $h\beta = (h/\bar{h})\bar{h}\beta = (h/\bar{h})f\gamma = (fh/\bar{h})f\gamma = (w/\bar{w})(\bar{f}/f)f\gamma = (\bar{f}/f)f\gamma = \bar{f}\gamma$.

This completes the proof.

Corollary 1. Any and all solutions of equation $x + y = z$ in SL_3Q such that $p(x) = p(y) = p(z) = t^3 - 1$ can be obtained by the following procedure:

(1) Take an arbitrary solution (a, w, r) in Q of the following equation:

$$(2a^3 + 3a^2 + a + 1)^2 - 4w((3a^2 + 3a + 1)^2/3 - (3a^2 + 3a + 1)w + w^2) = -3r^2;$$

Set $s = r(\rho - \rho^2)$.

(2) Set

$$p = ((2a^3 + 3a^2 + a + 1)^2 + s)/2,$$

$$q = ((2a^3 + 3a^2 + a + 1)^2 - s)/2;$$

$u = m + n\rho$, $v = m + n\rho^2$, where $m = (3a^2 + 3a + 1 - 3w)/3$, $n = (6a^2 + 6a + 2 - 3w)/3$;

(3) Select arbitrary b and c : $b \neq 0$, $c \neq 0$. Set $d = u/b$, $e = a\rho$, $g = v/c$, $f = p/(bg)$, $h = w/f$, $i = a\rho^2$; Set

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & 0 & \rho^2 \end{pmatrix}.$$

(4) Select an arbitrary $\alpha \in Q(\rho)$ such that $N(\alpha) = 1$. Set $\gamma = \bar{b}\alpha c^{-1}$, $\beta = \bar{\gamma}^{-1}$. Set $X = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & 0 & \beta \\ 0 & \gamma & 0 \end{pmatrix}$.

(5) Select any F such that $X = F(\bar{F})^{-1}$. One such F can always be chosen by Lemma 1.1. All such F form an open subspace of the linear Q -space determined by linear equations for entries' components (in the

form $k + l\rho$ of F (which correspond to the matrix equation $X\bar{F} = F$). The open subspace is determined by an additional condition: $\det F \neq 0$. Set $x = F^{-1}\Lambda F$, $y = F^{-1}AF$, $z = F^{-1}(A + \Lambda)F$. Then (x, y, z) is the solution.

Remark 1. The conjugation classes of obtained solutions are in one to one correspondence with solutions (a, w, r) of the equation in step 1.

Remark 2. To obtain all solutions, it is sufficient to attribute certain fixed values to b and c in step 3 and in step 4 instead of selecting b, c , and α arbitrarily. For example, all solutions could be obtained if we set $b = c = 1, \alpha = 1$.

Proof of Corollary 1. Since the polynomial $t^3 - 1$ is separable, any solution (x, y, z) can be obtained by a conjugation of a solution in the form $(\Lambda, A, \Lambda + A)$ for some matrix A over $\mathbb{Q}(\rho)$. This conjugation can be chosen to be induced by a matrix in $GL_3\mathbb{Q}(\rho)$. Thus, Theorem 1A.2 can be applied. We will use the notations of this theorem.

Conditions $2^0(b)$ and $2^0(c)$ can be considered as a system of linear equations in u and v . Condition $2^0(d)$, in conjunction with $pq = uvw$, reduces to a quadratic equation for p and q , from which p and q can be expressed through a and w . We substitute $r = (p - q)/(\rho - \rho^2)$ and obtain the equation in a, w , and r . From $2^0(e)$, it follows that a, w , and r are rational. All of this explains steps 1 and 2 of the procedure. Step 3 is self explanatory by definition of u, v, w, p , and q . (w, u and v defined in steps 1 and 2 are neither 0). Step 4 is determined by the proof of $2^0 \Rightarrow 1^0$ of Theorem 1A.2. Step 5 is due to the proof of Theorem 1.1 and the proof of Lemma 1.1.

Corollary 2. The equation $x + y = z$ has solutions in $SL_3\mathbb{Q}$ with $p(x) = p(y) = p(z) = t^3 - 1$.

Proof. By Corollary 1, it is sufficient to find solutions of the equation in the step (1) of the procedure. One such solution is $a = 0, w = 1, r = -1/3$.

Since our goal is to find (or to prove the existence of) solutions in $SL_3\mathbb{Z}$, we shall follow the procedure of Corollary 1 to find a solution in $SL_3\mathbb{Q}$ explicitly, so that we could then see if we can use a conjugation to transform it into a solution in $SL_3\mathbb{Z}$.

- (1) $a = 0, w = 1, r = 1/3; s = 1/3(\rho - \rho^2);$
- (2) $p = (1+s)/2, q = (1-s)/2$, i.e., $p = 2/3 + (1/3)\rho, q = 1/3 - (1/3)\rho; u = -2/3 - (1/3)\rho, v = -2/3 - (1/3)\rho^2;$
- (3) Set $b = 1, c = 1, d = -2/3 - (1/3)\rho, e = 0, f = \rho^2, g = -2/3 - (1/3)\rho^2, h = \rho, i = 0$. Set

$$A = \begin{pmatrix} 0 & 1 & 1 \\ \frac{-2-\rho}{3} & 0 & \rho^2 \\ \frac{-2-\rho^2}{3} & \rho & 0 \end{pmatrix}, \Lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & 0 & \rho^2 \end{pmatrix}.$$

$$(4) \alpha = 1, \beta = 1, \gamma = 1. X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

$$(5) F(\theta) = \theta I + \bar{\theta} X = \begin{pmatrix} \theta + \bar{\theta} & 0 & 0 \\ 0 & \theta & \bar{\theta} \\ 0 & \bar{\theta} & \theta \end{pmatrix}.$$

$$\text{Set } \theta = -\rho. \text{ We obtain } F = F(-\rho) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\rho & -\rho^2 \\ 0 & -\rho^2 & -\rho \end{pmatrix}, z = F^{-1} \Lambda F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}, y = F^{-1} A F = \begin{pmatrix} 0 & 1 & 1 \\ -\frac{1}{3} & 1 & -1 \\ \frac{2}{3} & 0 & -1 \end{pmatrix}, z = z + y = \begin{pmatrix} 1 & 1 & 1 \\ -\frac{1}{3} & 1 & -2 \\ \frac{2}{3} & 1 & -2 \end{pmatrix}.$$

1B. Applications to Theory of Symmetries on Affine Varieties

Theorem 1B.1. *On the affine variety over $\mathbb{C}$ determined by the equation $(2a^3 + 3a^2 + a + 1)^2 - 4w((3a^2 + 3a + 1)w + w^2) + 3r^2 = 0$ there exists a non-trivial symmetry of period 2 that commutes with the aw -plane symmetry and transforms rational points into rational points.*

Proof. The relationship of Remark 1 to Corollary 1 to Theorem 1A.2 can easily be established for complex solutions and matrices over $\mathbb{C}$. The commutation $(x, y, z) \mapsto (y, z, x)$ clearly induces an involution on variety of the statement of the corollary. It is easy to check that it is regular at every point; hence, it is a regular transformation, i.e., an automorphism. It is also easy to check (by explicit calculation, if not in any better way) that it maps $(0, 1, -1/3)$ into something different from $(0, 1, \pm 1/3)$, i.e., that it is different from the identical transformation and the aw -plane symmetry. From Remark 1 to Corollary 1 to Theorem 1A.2, it follows that rational points are transformed into rational points, q.e.d.

Remark 1. The above Theorem establishes an application of the conjugacy problem and equations in matrices to the theory of symmetries on affine varieties. A converse relationship may also be developed. Suppose that on the variety mentioned in the above Theorem, there exists a symmetry τ that transforms rational points into rational points, but can not be represented by polynomials with rational coefficients (the symmetry developed in the above Theorem may satisfy these properties)*. Then, we have a polynomial function $h(a, w, r)$ that takes on rational values at all rational points of the variety, but can not be represented on this variety by a polynomial with rational coefficients. Without loss of generality, we assume that h is minimal with respect to τ , i.e., that h is of at most first power with respect to τ . For any rational point x on the variety, let $v(x)$ be the vector whose coordinates are the values at x of all monomials that have a non-zero coefficient in h . Let the rank of the matrix consisting of all these vectors be s . Let $v(x_1), v(x_2), \dots, v(x_s)$ be linearly independent vectors, where $x_1, x_2, \dots, x_s$ are rational points on the variety. If N is the number of entries in every row, then $s < N$, since, otherwise, all coefficients of h would have to be rational since they ^{would} satisfy a regular system of linear equations with rational coefficients. By considering corresponding minors, we obtain that all rational points on the variety lie on a certain affine subvariety, determined by an additional equation of power no greater than 1 with respect to τ . The latter assertion, in conjunction with the fact that $(2a^3 + 3a^2 + a + 1)^2 - 4w((3a^2 + 3a + 1)^2/3 - (3a^2 + 3a + 1)w + w^2)$ is not always of the form $-3r^2$ with rational r if a and w are rational, implies that all rational points of the variety lie on a subvariety determined by an additional equation not involving r . According to Corollary 1 to Theorem 1A.2, this result would have immediate applications to the matrix equation and related conjugacy problems.

2. A Conjugacy Problem : Transformation of Families of Matrices Over $\mathbb{Q}$ Into Families of Matrices Over $\mathbb{Z}$

Theorem 2.1. (result A(i) in Section II above). If $A_1, \dots, A_m \in SL_n \mathbb{Q}$ (respectively $GL_n \mathbb{Q}, M_n \mathbb{Q}$), then the following conditions are equivalent:

- 1^0) There exists $F \in GL_n \mathbb{Q}$ such that $F^{-1}A_i F$ are integral matrices for all i ;
- 2^0) For any element C in the multiplicative semigroup G , generated by all A_i ($i = 1, \dots, m$), $\text{tr} C \in \mathbb{Z}$;
- 3^0) There exists an effectively obtainable integer d , such that $dC \in M_n \mathbb{Z}$ for any $C \in G$ (see 2^0);
- 4^0) There exists an effective algorithm for finding $F \in GL_n \mathbb{Q}$ such that $F^{-1}A_i F \in SL_n \mathbb{Z}$ (respectively $GL_n \mathbb{Z}, M_n \mathbb{Z}$) for all i .

Proof. The proof of the implication $1^0 \Rightarrow 2^0$ is trivial.

$1^0 \Rightarrow$ existence of d in 3^0 . The condition 1^0 is equivalent to the existence of a complete discrete lattice mapped into itself by the linear mappings corresponding to the given rational matrices. Since we can assume any single rational point (or vector) to be a lattice point, the question is equivalent to existence of d in 3^0 .

$2^0 \Rightarrow 3^0$. Let $M_\infty = \sum_{C \in G} \mathbb{Z}C$. We treat it as the limit position of M_i ($i \rightarrow \infty$), where $M_0 = \sum_{i=1}^m \mathbb{Z}A_i + \mathbb{Z}I$, and $M_{i+1} = \sum_{X,Y \in M_i} \mathbb{Z}XY$. Since the dimension of M_i quickly stabilises, 2^0 provides us with a sufficient number of equations for any $C \in G$ to find such d (see 3^0) effectively.

$3^0 \Rightarrow 4^0$. By 3^0 , M_i eventually gets sandwiched between 2 modules of the same rank, hence only finite number of "allowed" changes can occur as we pass from M_i to M_{i+1} starting from a point where the dimension is stabilised. Therefore, we can effectively find out whether the module will be stabilised, or a "forbidden" change will occur (no other alternatives exist). In the first case we will effectively find out what the ultimate module is. In the alternative case we will observe a "forbidden" change (in terms of the effectively found d). It is clear from the proof of equivalence of 1^0 and existence of d in 3^0 how the ultimate module M provides us with desired F .

$4^0 \Rightarrow 1^0$. The proof is trivial.

Corollary 1. If $A, B \in SL_2 \mathbb{Q}$, then the following conditions are equivalent:

- 1^0) There exists $F \in GL_2 \mathbb{Q}$ such that $F^{-1}AF, F^{-1}BF \in SL_2 \mathbb{Z}$;
- 2^0) $\text{tr} A \in \mathbb{Z}, \text{tr} B \in \mathbb{Z}, \det(A+B) \in \mathbb{Z}$.

Proof. As $\det(A+B) = \det A + \det B + (\text{tr} A)(\text{tr} B) - \text{tr}(AB)$ condition $\det(A+B) \in \mathbb{Z}$ in 2^0 can (equivalently) be replaced by the condition $\text{tr}(AB) \in \mathbb{Z}$. 2^0 thus implies that A, B , and AB each satisfy some monic equation of degree 2 with integer coefficients. Therefore, any $C \in G$ (see Theorem), which has AA, BB , or $ABAB$ in its representation as the product of A^* and B^* , can be represented as the sum of elements of G which are the products of a lesser number of A^* and B^* . Therefore, to derive condition 2^0 of the theorem, it is sufficient to prove that traces of the following matrices are integers: $A, B, AB, BA, ABA, BAB, BABA$. This follows immediately from the fact that traces of A, B , and AB are integers, and so are traces of their inverse, as their determinants equal 1. Thus, by the theorem, 1^0 follows from 2^0 . The converse is obvious.

2A. Applications to Fermat's Equation in Matrices

Theorem 2A.1. For any solution $x^n + y^n = z^n$ in $SL_2\mathbb{Q}$ there exists a matrix f in $GL_2\mathbb{Q}$ such that for $x_1 = f^{-1}xf$, $y_1 = f^{-1}yf$, $z_1 = f^{-1}zf$, we have $x_1^n + y_1^n = z_1^n$ and $x_1^n \in SL_2\mathbb{Z}$, $y_1^n \in SL_2\mathbb{Z}$, $z_1^n \in SL_2\mathbb{Z}$, $x_1 \in SL_2\mathbb{Q}$, $y_1 \in SL_2\mathbb{Q}$, $z_1 \in SL_2\mathbb{Q}$.

Proof. This follows immediately from Corollary 1 of Theorem 2.1

Theorem 2A.2. Equation $x^n + y^n = z^n$ has periodic solutions in $SL_2\mathbb{Z}$ for any n not divisible by 3.

Proof. We start with the solution

$$x_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}, y_0 = \begin{pmatrix} 0 & 1 & 1 \\ -\frac{1}{3} & 1 & -1 \\ -\frac{2}{3} & 0 & -1 \end{pmatrix}, z_0 = \begin{pmatrix} 1 & 1 & 1 \\ -\frac{1}{3} & 1 & -2 \\ -\frac{2}{3} & 1 & -2 \end{pmatrix} \quad (n \equiv 1 \pmod{3})$$

in $SL_3\mathbb{Q}$ derived in Corollary 2 of Theorem 1A.2. By following the algorithm of 4° in Theorem 2.1, we obtain

$F = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix}$, which leads to the solution in $SL_3\mathbb{Z}$:

$$x = F^{-1}x_0F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 3 & -2 \end{pmatrix}, y = F^{-1}y_0F = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 1 & -1 \\ -2 & 0 & -1 \end{pmatrix}, z = x + y = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 2 & -2 \\ -2 & 3 & -3 \end{pmatrix}.$$

IV. Related Open Questions and Directions for Further Research

Invariant spaces (i.e. the spaces mapped into themselves under certain linear mappings) might be used to resolve the question regarding the existence of conjugation, transforming the given set of non-rational matrices into rational matrices. Although initial vector (or point) now matters, this approach might still lead to conditions different from those derived in the paper and dealing with norms. How might they be compared with each other to obtain further insights and results?

For matrices of the order greater than 2, is it possible to find an effective limit m such that consideration of products of up to m matrices A_i is sufficient in IIA(i) 2°? ($m = 2$ works for 2 matrices 2 by 2.) Can we find other convenient ways to utilise the integral origin of the matrices in irrational case?

Still unsolved is $x^3 + y^3 = z^3$ in $SL_3\mathbb{Z}$ and $GL_3\mathbb{Z}$, as well as in $SL_3\mathbb{Q}$. The problem has to do with ranges of norms but in much more complex form than in conditions IIA(iii) 2° and IIA(iv) 2°, especially when we have to deal with non-cyclic extension. (In case of non-cyclic extensions, Theory of Galois cohomologies should substitute Hilbert's Theorem #90). How about solutions in $SL_m\mathbb{Z}$ for larger m ? From this paper it follows that periodic solutions exist for all exponents for which solutions exist in $SL_2\mathbb{Z}$.

In section 1B, we found on a certain affine variety over $\mathbb{C}$ a new symmetry transforming rational points into rational points (Theorem 1B.1). Many others can be derived from the connection between certain equations in matrices and certain equations in various fields. How can they be used together with classical Number Theory and Algebraic Geometry approaches to find complete set of symmetries on these varieties or to obtain infinite families of solutions from one? If the coefficients of the polynomial representing one of

these symmetries are not all rational, then all rational points lie on a certain subvariety; this result would have immediate applications to the original matrix equation (see Remark 1 to Theorem 1B.1).

Since Theory of Symmetries is closely related to Dynamical Systems, which in turn is widely applied in Economical and other Sciences, new results concerning symmetries on affine varieties may lead to new insights in these fields.

In general, the relationship established in this paper between the conjugacy problem, equations in matrices and theory of symmetries on affine varieties may lead to a number of new results in each of this fields and should be further researched.

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The Conjugacy Problem for Families of Matrices, With Applications to Fermat's Equation in Matrices and Theory of Symmetries on Affine Varieties

BY ALEXANDR KHAZANOV

(an alternative version of the previous article)

What Matrix Equations and Conjugations Have to Do With Symmetry

IMAGINE that you've just drawn a four-dimensional figure in a six-dimensional. Your picture looks weird; yet, you believe it's beautiful. Others ask you for rationale. And so you naturally begin seek symmetries, the major attribute of beauty. Your figure is so complex (incidentally, it is a complex affine surface in a three-dimensional complex affine space), that you naturally try to associate it with something simpler and more obviously symmetric. With matrix equations involving symmetric polynomials, for instance. Or better yet, with the conjugation classes of solutions to such equations. This requires some insights into matrix equations.

Solving Matrix Equations

You are working on an equation in a 3×3 matrix. Your first idea is to substitute variables for each entry and to write out the original matrix equation as a system of equations in these variables. However, you are soon disappointed by this idea: the number of variables involved is 27, and the equations you obtain in these variables are impossible to handle. And at once you recall Thoreau's "Simplify, Simplify." "Simplify what?" you ask yourself. Matrix equations are very difficult to simplify because there isn't much room for action: you can't even switch numbers around since $xy \neq yx$. "So what can I simplify?" you ask yourself again, this time almost desperately. Suddenly, the answer comes to you: if not equations, then solutions. Your plan slowly takes shape: to associate with any solution a "simple" one; to find all "simple" solutions; and to find all solutions associated with any "simple" solutions.

Conjugations: Many Solutions From One

You have realized that you need a procedure to associate many solutions with one. You recall that a reflection of a nonsymmetric object preserves all its metric properties, yet it changes in orientation and outward appearance. Matrix multiplication is in some sense nonsymmetric (more precisely, noncommutative): $xy \neq yx$. This provides for a kind of reflection: $f^{-1}fx \rightarrow f^{-1}xf$. This "reflection," called conjugation, changes the outward appearance of matrices and can make them "simpler"; yet, it preserves their fundamental algebraic properties; in particular, solutions to any equation are mapped into solutions to the same equation.

The Conjugacy Problem

According to a standard result, any matrix satisfying certain technical conditions can be transformed by a conjugation into a diagonal matrix, i.e. a matrix all of whose nonzero entries lie on the main diagonal. This result enables us to complete the first part of our plan of solving some matrix equations: the solutions in matrices of which one is diagonal will be the "simpler" solutions in that plan. Finding all such "simple" solutions is a technical task that we won't describe. The much more interesting problem is that of finding all solutions of the desired kind (e.g. with rational or integer entries) in the conjugation class associated with any such "simple" solution. This problem is called the conjugacy problem.

Conjugacy Problem: Transformation Into Matrices With Rational Entries

You are given a family of matrices with some complex elements. You are asked to either find a conjugation that simultaneously transforms all these matrices into matrices with rational

elements, or prove that such conjugation doesn't exist. It would be helpful to interpret the rationality of the conjugate matrices' elements as a system of equations in the matrix inducing the conjugation. But how can this be done? Inspiration doesn't come to you; therefore, you begin to paraphrase the problem, trying to find a formulation that would best illuminate its nature.

What are you really trying to do? You are trying to reduce the condition that certain matrices have rational elements to a system of equations. In other words, you are trying to find a system of equations satisfied by matrices with rational elements and only by them. What kind of a problem is this? This is a problem of identification of numbers by equations. The problem seems very difficult in general; therefore, you decide to consider some simple special cases.

Any rational number can be identified by a linear equation with integer coefficients satisfied by it. Some irrational numbers can also be associated with polynomial equations with integer coefficients satisfied by them. It occurs to you that it may be interesting to compare the patterns of identification of rational and irrational numbers by polynomial equations: after all, part of the problem is to find something that distinguishes rational numbers from all others. And, wow, you find the desired fundamental difference: whereas a rational number is defined uniquely by a linear equation with integer coefficients satisfied by it, no system of polynomial equations with integer coefficients would identify an irrational number uniquely.

It seems like you've gotten some real progress: you've found a property that distinguishes rational numbers from all others. But how can you reduce this to an easy-to-handle system of equations? The answer lies in the obvious relationship between uniqueness and equality. If a property defines a number uniquely then any two numbers satisfying the property are equal, and vice versa. Since equations with integer coefficients satisfied by a specific number c define this number c uniquely if and only if it is rational, rationality of number c is equivalent to its equality to any number satisfying the same set of polynomial equations with integer coefficients. Due to a technical result, this is equivalent to the system of equations $c = \delta(c)$ where δ runs through all one to one transformations of the field of complex numbers mapping sums into sums and products into products. (Note that such transformations preserve the fundamental algebraic properties of any family of numbers: for example, solutions to any equation are mapped into solutions to the same equation). These transformations, called automorphisms, can be naturally extended to all square matrices (elementwise); the condition that a matrix H has rational elements is then equivalent to the system of equations $H = E(H)$ where δ runs through all extended automorphisms of the field of complex numbers.

Now you have reduced the conjugacy problem to the problem of finding all matrices satisfying some systems of equations. Well, in a sense you're back to where you started: solving matrix equations. On the other hand, there are some reasons for you to hope that your new equations are easier to handle: they involve only one variable, F , the matrix inducing the conjugation. Furthermore, if you substitute $X_\delta = F(\delta(F))^{-1}$, the equations become linear of the form $AX_\delta = X_\delta \cdot \delta(A)$. The only nontrivial question left for you to answer is that of finding conditions for a family of matrices (X_δ) some matrix F by formulas $X_\delta = F(\delta(F))^{-1}$. You soon recall a very similarly solved question in Galois Theory, and realize that a very similar approach based on the generalized notion of first Galois Cohomology works to resolve the new problem you faced.

In Conclusion

Now you can implement all parts of your plan of solving matrix equations in matrices with rational elements. You are also ready to discover the nontrivial symmetry on your figure. But

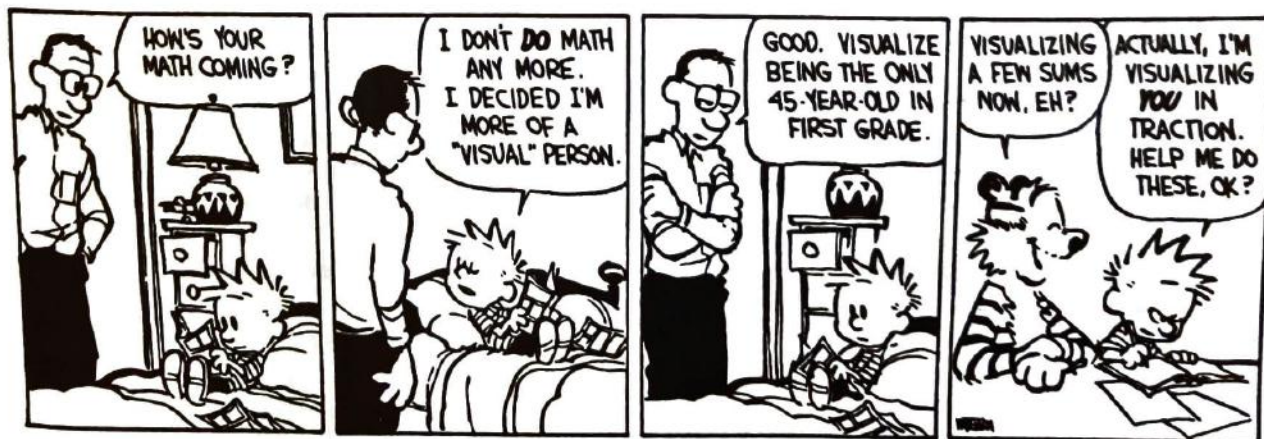
you feel inspired, and, rather than going through the routine of implementing procedures, you choose to try solving the conjugacy problem for transformations into matrices with integer elements. You solve this problem using the relationship between matrices and linear transformations, conjugations and changes of coordinates, wholeness of matrices' elements and invariance of the coordinate lattice under corresponding transformations.

You have developed a great deal of ideas and are now in the mood to sit down in front of a table and to implement them technically.

Here are some of the results you have obtained after long hours of work. Above all, you have established a relationship between the conjugacy problem for families of matrices, matrix equations, symmetries on varieties, and representation of numbers by norms of elements in some fields (the latter comes from the conditions you have developed for the conjugacy problem). More specifically, you have established a one to one correspondence between the conjugation classes of solutions of the system $x^3 = y^3 = (x+y)^3 = I$ in matrices with rational elements and rational points (a, w, r) on the affine surface defined by the equation.

Using this relationship and observing that transposition $(x, y) \rightarrow (y, x)$ maps solutions to the above system of matrix equations into solutions to the same system of equations, you have obtained a nontrivial symmetry on the affine surface defined by the above equation in a, w, r , the hidden reason as to why it looked so special. You have also shown that Fermat's equation $x^n + y^n = z^n$ has solutions in square matrices of determinant 1 of any size larger than 2 with integer elements as long as n is not a multiple of 3. You have shown that it doesn't have such solutions in matrices 2 by 2 if n is a multiple of either 3 or 4, and has them for all other n . You have solved several special cases of the conjugacy problem.

Alas! It is time to say good bye to the wonderful world of matrices, symmetries, conjugations, varieties. So much work was left behind while you were absorbed in your adventures. But in the bitter moments of parting, you also say hopefully: "See you later."



The LAN Collection of Favorite Proofs, Limericks, and Anecdotes

BY IRIS LAN

PLUS ONE OR MINUS ONE?

$$\begin{aligned}
 -1 &= -1 \\
 -1/1 &= -1/1 \\
 -1/1 &= 1/-1 \\
 \text{square root of } (-1/1) &= \text{square root of } (1/-1) \\
 (\text{square root of } -1)/1 &= 1/(\text{square root of } -1) \\
 i/1 &= 1/i \\
 i * i &= 1 \\
 -1 &= 1
 \end{aligned}$$

"PROOF" THAT TWO EQUALS ONE (2=1)

$$\begin{aligned}
 \text{Let } x &= y. \\
 \text{Then } x^2 &= xy, \\
 \text{so } x^2 - y^2 &= xy - y^2, \\
 \text{or, in other words, } (x + y)(x - y) &= y(x - y) \\
 \text{Cancelling yields } x + y = y, \text{ hence } 2y = y \text{ and } 2 &= 1.
 \end{aligned}$$

LET Y = NUMBER OF X IT TAKES TO CHANGE A LIGHT BULB

<u>X</u>	<u>Y</u>
Real functions	None, it's too complex for them
Numerical Analysis	0.99987, after four iterations
Statisticians	One, with 0.95 % confidence - or one, on the average
Constructive mathematicians	None. They do not believe in infinitesimal rotations
Number theorists	Only one, but the margin is too small for proof.
Classical geometers	None. It cannot be done with a straight edge and compass

MATHEMATICIAN'S NICE CLEAN LIMERICK

$$\int_1^{\sqrt[3]{3}} z^2 dz \times \cos\left(\frac{3 \times \pi}{9}\right) = \ln(\sqrt[3]{e})$$

TRANSLATION OF MATHEMATICIAN'S NICE CLEAN LIMERICK

*Integral z-squared dz
 From 1 to the cube root of three
 Times the cosine
 Of 3 pi over nine
 Equals log of the cube root of e*

MATHEMATICIAN'S FAREWELL LIMERICK

A mathematician of Trinity
 Was computing the cube of infinity
 But the number of digits
 Soon gave him such fidgets
 He dropped it and took up divinity.

MATHEMATICIAN'S NIGHTMARE LIBRARY

First Aid for Dedekind Cuts
Jacobians and the Struggle for Independence
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FROM THE GREETING CARD SECTION OF THE MATHEMATICIAN'S NIGHTMARE BOOKSTORE

Sorry I reformatted your disk!
 So---you were denied tenure.
 Thank you for your flawed proof of my theorem.
 Happy 1011001th birthday!
 And you thought Hilbert had problems?

A MODEL ANSWER YIELDS SOME RELATIVE TRANQUILITY

Einstein in America found himself engulfed by fame. Autograph seekers pursued him everywhere. A chance remark he made during a walk - "Rafiniert ist der Herr Gott, aber boschaft ist er nicht" (God is subtle but not malicious) - wound up engraved in marble over a fireplace at the Princeton math department. He had to abandon his magnificent ground-floor office there because sightseers kept peering through his window.

A quiet scholarly man, Einstein was sometimes overwhelmed by this relentless public attention. And all the while, his face grew more and more famous because of the many patient hours he spent posing for refugee artists who needed the money by creating yet another Einstein portrait.

One day, Einstein was riding the train from New York to Princeton, when his reverie was interrupted by an inquisitive stranger who said, "I don't think I know you - but your face is so familiar...How could that be?"

Einstein said wearily, "Perhaps because I spend my days as an artists' model."

"Ah," said the man, "that's why I know you face!" And he left Einstein in peace for the rest of the trip.

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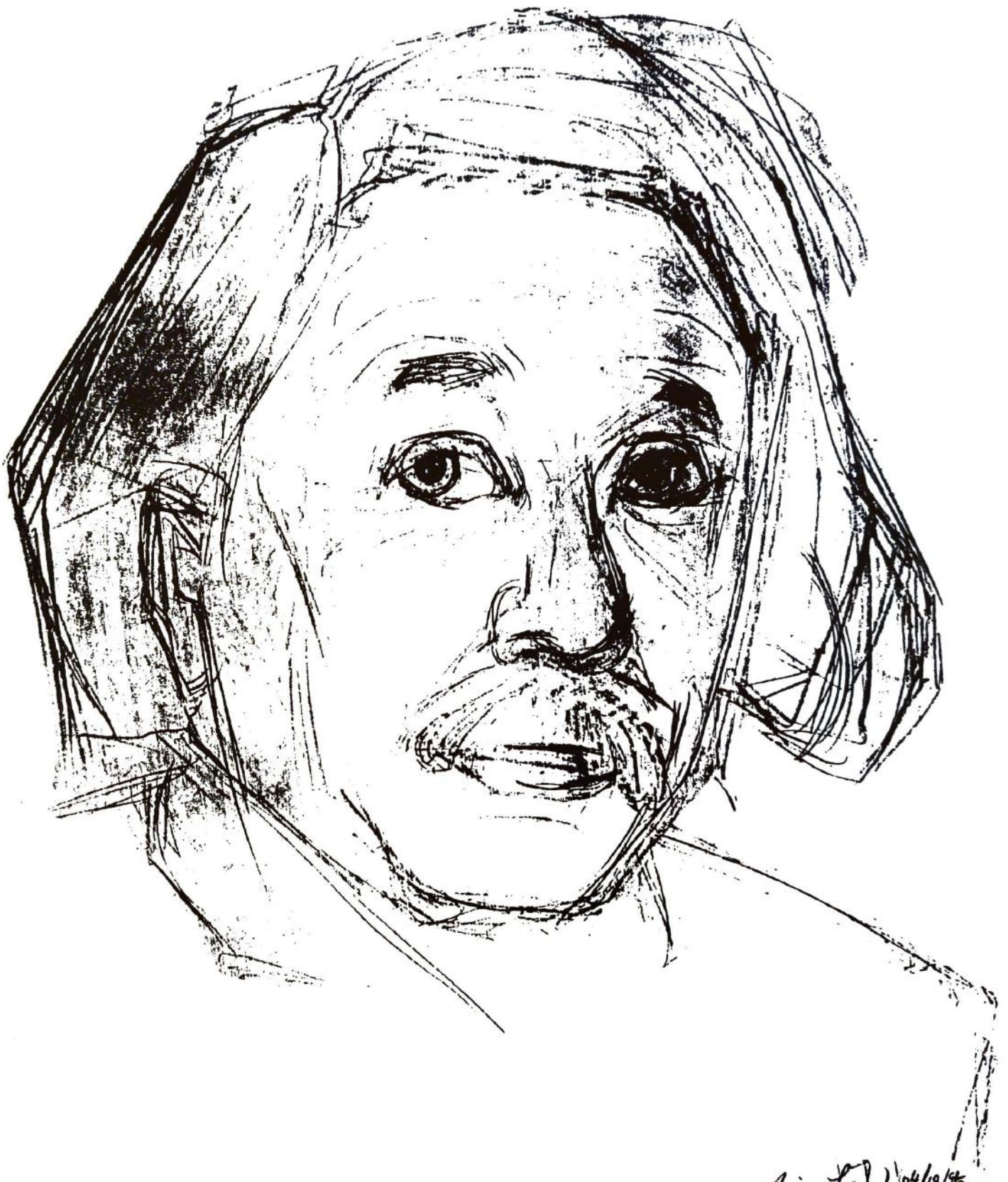
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Jess Law 04/19/15



The Two Brothers

A Saxon king, displeased with the greed shown by his two sons, left a will with an unusual provision: The sons were to mount their horses in a tiny town on the border of the kingdom and ride, without dismounting more than once, to the gate of the king's castle. The son whose horse arrived *second* at the castle gate was to be awarded the entire fortune.

When the king passed away, the sons began their "race," moving along slowly together for days on end. Each tried to go slower than the other and after a while, they got so sleepy they had to dismount and get some sleep at an inn. While they slept, they each had their footmen alerted to notify them when the other left the inn. Actually, they left the inn together. As they were about to mount their horses, one brother whispered a few words to the other, they both laughed, jumped into the saddles and raced the horses as fast as they could to the castle gate.

It doesn't matter to us which brother won the fortune, but can you figure out what the brother whispered?

Solution:

The brother simply said, "Let's swap horses!" The king's will stated that the son whose horse arrived second would get the fortune. Therefore, by riding the other one's horse, the winner of the race could be sure his own horse would be second at the castle gate. So they swapped and sped off.

Mathematics in Origami and Paper Folding

BY JERRY LEE

ORIGAMI, the centuries-old art of folding paper into decorative objects, is believed by historians to have originated in China. However, it was in Japan that origami flourished and was incorporated into the culture under the five rules established by Husimi. These rules stated that: a) one starts with a square sheet of paper; b) one forbids cutting or gluing; c) the paper must be flat at the end; d) only straight folds are allowed; and e) one mustn't ignore the physical qualities of the paper. This art has come a long way since Husimi's time with the recognition of the close tie that exists between mathematics, particularly geometry, and origami. In fact, two of the existing five categories of origami concentrate on the creation and study of polyhedra (including stellated) and crease patterns that tile the plane; they are known as modular and tessellation origami. Of particular interest are some of the properties inherent in the crease-pattern of flat origami and the uses of origami and paper folding in geometry.

In geometry, how does one manage to trisect an angle without instruments (i.e. rulers, protractors, etc.) and only a piece of paper? It seems rather difficult using only a sheet of paper. With origami paper which is perfectly square, one can trisect a right angle merely by folding the paper (Figure 1). The geometric proof uses the properties of 30° - 60° - 90° triangles (Figure 2).

How about dividing a square into any number of divisions with only origami paper? Before answering, one should realize that it is quite easy to divide the sides of a square into 2^n divisions where n is a positive integer (halves, quarters, eighths, etc.) by folding the paper in half n times. However, it is quite difficult to divide the paper into an odd number of divisions without instruments, but there are ways of dividing a square into thirds and fifths, as well as a general method for obtaining any number of divisions.

Figure 1

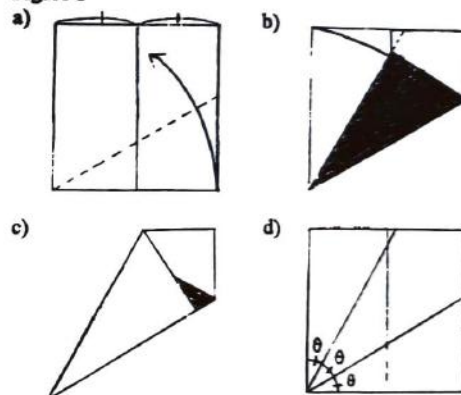


Figure 2

a) From figure 1b. Let x = the length of the side of square.

Using the Pythagorean theorem

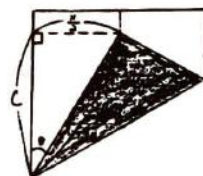
$$(x/2)^2 + c^2 = x^2$$

$$x^2 - (x/2)^2 = c^2$$

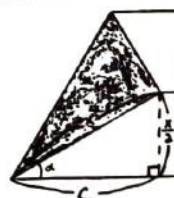
$$3x^2/4 = c^2$$

$$(x\sqrt{3})/2 = c$$

A right triangle with legs a and b where $a:b = 1:\sqrt{3}$ is a 30° - 60° - 90° triangle with the 30° opposite the leg of length a . $m\angle\theta = 30$



b) One only needs to repeat this process with the opposite corner to show the trisection.



Use Pythagorean theorem

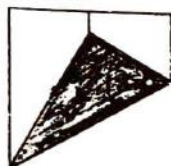
$$c = (x\sqrt{3})/2$$

$$\therefore m\angle\theta = 30$$

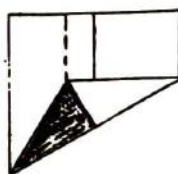
To divide a square into thirds, the process mentioned above to trisect an angle can be employed.

Figure 3

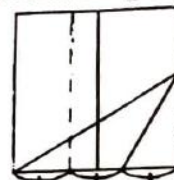
a) From Fig. 1b



b)



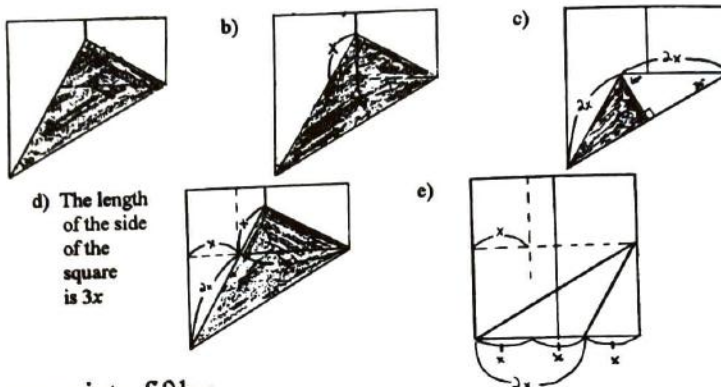
c) seen after folding along dots in b)



The proof of this requires the use of 30° - 60° - 90° triangle again:

Figure 4

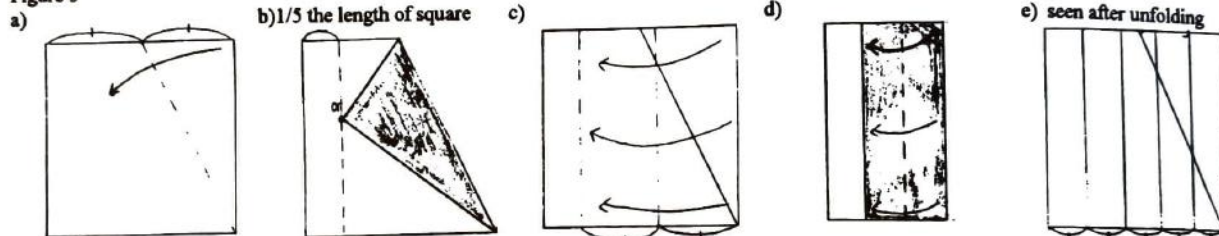
a) From Fig. 1b, angle measurements were already established



In a $30^\circ-60^\circ-90^\circ$ triangle, the length of the hypotenuse is always twice the length of the leg opposite the 30°

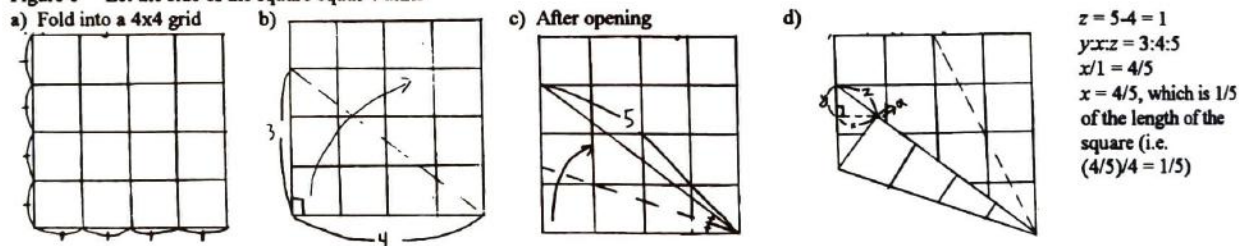
To divide a square into fifths:

Figure 5



The proof of this requires the use of the 3-4-5 right triangle and a 4×4 grid.

Figure 6 Let the side of the square equal 4 units



The general method of making divisions breaks the rules of a bit by using a "ruler" of sorts. This so-called "ruler" will be made by folding a sheet of origami paper into an even number of divisions. One can then use this to divide other sheets of origami paper into any number of divisions, as shown in Figure 7. Note that the "ruler" does not need to have eight divisions as Figure 7 does. This was done strictly to provide an example. Also, the other sheets of paper that one wishes to divide should be positioned on the "ruler" so that one of the vertices ("a") coincides with one of the "ruler's" so that the adjacent vertex ("b") lies on the desired numbered fold of the "ruler." For example, if one wishes to divide the side of a square into seven divisions as in Figure 7, one would line up vertex "b" on the seventh fold. Additionally, if one is even in need of determining whether or not angles have measurements of 30, 45, 60, 75, 90, and 120, one can consult Figure 8a and Figure 8b.

Figure 7

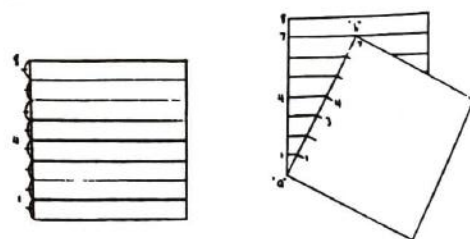
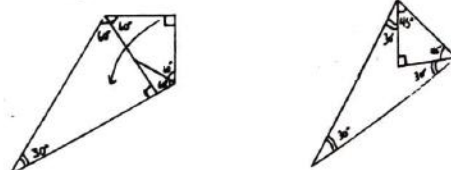


Figure 8

a) From Fig. 1c

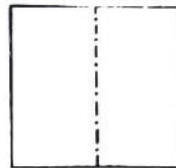
b) Fold a)



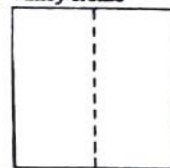
Now, on to the crease-patterns. The crease-patterns are simply the creases that one sees on the origami paper after folding. Although there are an infinite-number of crease-patterns, there are only two types of creases: mountain creases, which are represented by a dot-dash-dot line, and valley creases, which are drawn as a dashed line (Figure 9).

Figure 9

a) Mountain crease



b) Valley crease



Before I can describe some of the local properties of the crease-patterns of a flat origami, let us mathematically define origami by the pair (C, f) where C is a set of lines in $[0, 1] \times [1, 0]$, which defines the creases, and f is a function where $f: C \rightarrow (-\pi, \pi)$ which describes how each crease is folded (how far, in what direction). The pair (C, f) defines a function from the square $[0, 1] \times [1, 0]$ of $\mathbb{R}^2$ to the actual realm of origami of $\mathbb{R}^3$. Note: In order to guarantee that the function does not include folds which require the tearing of the paper, the function must be 1 to 1.

Definition: An origami is a pair (C, f) , where C is the crease set and $f: C \rightarrow (-\pi, \pi)$, such that the mapping of (C, f) produced from $[0, 1] \times [1, 0]$ into $\mathbb{R}^3$ is 1-1.

Definition: A flat origami is an origami such that if we take the limit as each valley crease is folded to π radians and as mountain crease is folded to $-\pi$ radians, then the origami is still 1-1 as we take the limit.

The first property of flat origami is that the difference between the number of mountain creases and valley creases is 2 (Maekawa's theorem).

Proof (by Jan Siwanowicz): Consider the unit disk D after it is folded. The one vertex in the crease-pattern will then be a point in this folded figure. If we cut this point off, a polygonal cross-section of the folded paper is revealed. Notice that since this is a flat origami all the interior angles of this polygon will either be 0 or 2π radians.



Now, the mountain folds M of C (crease set) correspond to the 0 angle of the polygon, and the valley folds V correspond to the 2

π angles. If we let n be the number of creases (which equals the number of sides of our polygon) in C , then the sum of the interior angles of the polygon equal $(n-2)\pi$. But since $n = M+V$ (the total number of creases) as well, we have

$$0M + 2\pi V = (M+V-2)\pi$$

When manipulated, one obtains $M-V = 2$. Q.E.D.

But if this is so ($M-V = 2$), then the number of creases surrounding the vertex is always even because the difference between the M and V creases is equal to 2. This provides us with a corollary to Maekawa's theorem.

Corollary: The number of creases surrounding a vertex is even.

There exists a close tie between origami and mathematics; origami can contribute to the field of mathematics and vice versa. Origami should, therefore, be recognized as a mathematical tool and should be explored even further.

Fractal Dimensions

BY CHE KING LEO

FRACTALS represent the basis of much of nature, but they are a new branch of mathematics, the elements of which were unknown until about one hundred years ago. An important concept dealing with fractals is fractal dimension. To get an idea of fractal dimension, consider that in nature, most objects are not exactly three dimensional. For example, mountains on a plain are three dimensional in places where mountains exist, but are only two dimensional on the plains. What dimension would the entire region have? The true dimension is in between two and three, and there is a special way to find it.

Consider a line (one dimension), a square (two dimensions), and a cube, divided into equal segments of $1/n$, with n pieces in the line, n^2 in the square, and n^3 in the cube. Using logarithms, we find that $\log(\text{number of pieces in the line}) = \log n' = 1(\log n)$. Similarly in the square, $\log(\text{no. of pieces in the square}) = \log n^2 = 2(\log n)$, and in the cube, $\log(\text{no. of pieces in the cube}) = \log n^3 = 3(\log n)$. Since fractals are all self-similar (meaning that the fractal resembles itself at different levels of magnification), we find that a magnification of a piece by n times would give us the original object. So, we can compute dimension D :

$$D = \log(\text{number of pieces}) / \log(\text{magnification})$$

In the case of the line, $D = (\log n')/(\log n) = (\log n)/(\log n) = 1$. For the square, $D = (\log n^2)/(\log n) = 2(\log n)/(\log n) = 2$. For the cube, $D = (\log n^3)/(\log n) = 3(\log n)/(\log n) = 3$. These values correspond to one, two, and three dimensions respectively, as we would expect.



Fig. 1 Sierpinski triangle

A common fractal is the Sierpinski triangle, which is generated by removing middle triangles indefinitely (Figure 1). Magnifying the fractal two times yields three pieces. Therefore, for that magnification, the dimension $D = (\log 3)/(\log 2) = 1.58 \dots$. This makes sense, because it's not quite two dimensional but it is definitely not flat.

Another fractal is the Koch snowflake (Figure 2). Beginning with a line, every side is changed to: , where all four sides are of equal length and each new side is $1/3$ the length of the original side (i.e., an equilateral triangle is raised from every side). Thus, if the fractal is magnified three times, there are four pieces. Therefore, $D = (\log 4)/(\log 3) = 1.26 \dots$. As in the Sierpinski triangle, the dimension is between one and two.

Another fractal is the Koch snowflake (Figure 2). Beginning with a line, every side is changed to: , where all four sides are of equal length and each new



Fig. 2 Koch snowflake

A good way of perceiving fractal dimension in nature is a piece of aluminum foil. Fresh off the roll, the foil has a dimension slightly over two. Crumpled and opened, the dimension changes to just under 3.

Ramblings of a Famous Mathematician Gone Crazy

BY CAROL KIM

1. NUMBERS are jumping around inside my head. Equations galore: Where to begin? Calculus stinks. 5, 2, 8, 1, 9, 0, 6, 4, ...#####! Hate formulas. Love the number seven ... it's a calm yet cheerful number. Where to stop? Find the answer to the following:
 $1994 + 1998 + 2002 + 2006 + \dots + 51989504$
 Hint: Use $s = (n/2)(a_1 + a_n)$ and $a_n = a_1 + (n - 1)d$. What?!?

The ultimate question: "Find the value of n such that n plus everything in your life makes you feel all right. What would n equal? Solve for n ."

Can you do it?

2. The color ... fold. Can hurt, yet oh how pretty it is! I do not want to throw it out because it is worth so much, but it is so demanding having to constantly take care of it. Sometimes I even conceal it, just in case it gets hurt or damaged. But you want to look good, so you take the chance. Nike. Just do it.
3. I met Sisypheus today. He's a one-of-a-kind. Cool guy, but he always tortures himself endlessly trying to complete some task. Maybe it's calculus. Recall recent relating ramble. Ready?
4. I have so much homework. I talked to Sisypheus. He said, "There's no sun without shadow. It is essential to know night." Thanks a lot. Read. Reflect. Realize. Reason. Resolve. See? Confusion can be conquered. Be aware.
5. Calculus is very hard. Sometimes I blame the mathematicians for formulating the complex equations and derivations to unravel what seems like simple questions. But it is not their fault that I do not understand. Tests are arduous. I wish I could change some of the mistakes I made on them. Oh well. Gripe. Gripe.
6. Life. Live it. If it is not put to the test, it is not worth living.
7. What a nice sunny day! I feel like taking a walk. I need new shoes. I hate breaking them in, though. Sometimes I do not even want to bother. But I cannot go on with these old ones (they cannot be repaired). I am going to buy a nicer pair of shoes this time. Once I get used to them, I will feel a great deal better. What would be the best solution? Should I actually buy new ones? What should I do with the old? Keep them? Will I outgrow these? Should I get a size six or a six and a half? Do I really want to break these in? Do I like them? Just wear the shoes. Be happy.

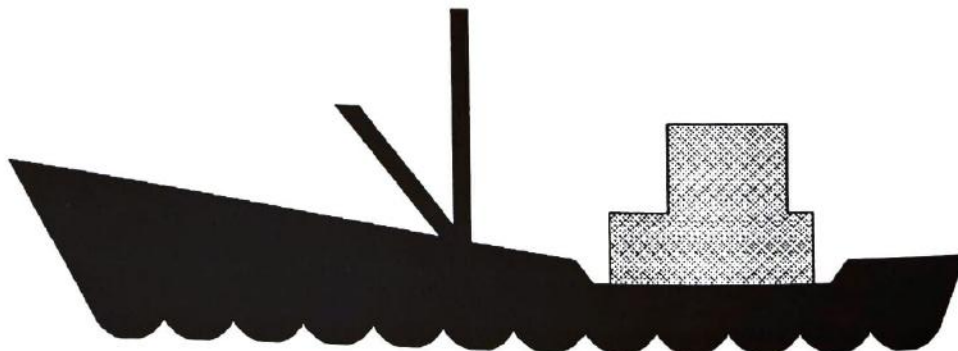
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Stuyvesant Math Team: 1994-1995

BY BIFF HENDERSON

THIS class of mathteamers has had a few diverse, not to mention brilliant, students. This year's crop included National Knitting Competition first prize winner Zvi Mowshowitz, prize mogul skier Boris Granovskiy, and Guinness Book of World Records holder for balancing on a streetlight Daniel Stronger (17 hours, 53 minutes). Each one of these is a success story in their own right, and will be discussed as such.

Mowshowitz was born deep in the heart of Athens, Greece, but moved to New York at the age of three. Living in the depths of Brighton Beach did not exactly fit the mold of previous knitting champions, but Zvi had an uncle who the champion in Greece. "He was a great spiritual influence on my life. He taught me how to knit at the age of seven, when, due to an unfortunate asthmatic condition, I was forced to give up playing street hockey." Journeying to Topeka, Kansas in March for the nationals, Zvi was able to soundly defeat the previous year's winner, Ranjith Rajagopalan of Tucson, Arizona, by a cumulative five-judge score of 48.95 to 47.50. Says Knitting Teammate Philip DeAngelis: "Zvi is amazing. He can knit a regulation rug in fourteen seconds easily." Zvi's best time is 17.83. Zvi Mowshowitz is now the hero of his birthplace of Athens, and May 5 has been officially declared Zvi Day.

Boris Granovskiy grew up in the depths of the city, at the corner of Madison and 45th. At age 4 he learned how to ski, and guided by his parents, Winston and Estelle Granovskiy, he placed fourth in the downhill in the Under 12's at age 10. However, when he went over his first mogul at age 11, he knew he had a long future ahead of him. "I tried to steer him away from the dangerous world of moguls, but the wild kid couldn't be held back. A father's heart flutters whenever his kid goes in the air. I've taken out insurance policies on both his legs. I feel safer now, but not a day goes by where I don't think about it," says Winston Granovskiy. This year, at age 15, Boris placed third in the Under 18 moguls state competition in Cornwall, NY behind Ron Walker of Canajoharie and Scott Crosby of Rochester. Boris says of the Stuy math team, "No big egos here. Knitting champion, spy, truck driver, we all leave our egos at the door." His mentor and now Math Team captain Elliot Shmukler, nods in quiet approval.

Professional truck driver Josh Berne got his start when a friend asked him to care for a Monster Truck at age 10. This gave him an international ranking of #4913, and by age 12, Josh had taken 119th place in a national show, his first time in a truck, which jumped his rating to #476. By age 14, he had cracked the top 100, and his current rating is 3rd in the world, the youngest Top Three driver ever. His breakthrough event was the 1994 Bam Bam Show, where, driving his own contraption, a car that runs solely on Jolt Cola, he took first place by setting a new world record of 16 cars. He bettered that last December, after India's Mark Goodman had tied it, and now holds the record of 17 cars. However, Josh says, "My first devotion is to the Math Team. They're a great bunch of guys, and I wish them the best of luck in life." Josh also claims the most important event in his life was not the 1994 Bam Bam Show, but when he was a member of the winning Stuyvesant Math Team Junior teams for both terms in 1993-4, the only student ever to do so. Says manager and a current Math Team captain Hoi Wong, "Josh is amazing. He never sleeps, never eats and still blows them away."

Daniel Stronger found his balance at age three, when he was the only member of his preschool class who could stand on top of the ten-foot fireman's pole without falling-on his nose. Much to the horror of his parents, Roz and Philip Stronger of Little Neck, he did not outgrow this

phase, standing on trees at age five, and he would always "go up on the roof to fetch the frisbees," says longtime friend and now fellow Math Team member Mike Develin. "He wasn't particularly strong, or smart, or quick, or handsome, but he could outmaneuver us all in the touch football games, and doggone it, people liked him," adds another friend and now fellow Math Team member, Bobby Woods-Corwin. Stronger himself claims, "I didn't even know about Guinness until [Math Team captain Yuditsky] Leonid jumped into the air and pumped his fists, yelling 'Yeah!' I was just trying to catch the bird that kept running away with the 9 of Spades." In true Math Team spirit, Daniel set the record at ARML last year, which the New York City A Team won handily.

By the way, the Stuyvesant Math Team did well in all competitions.

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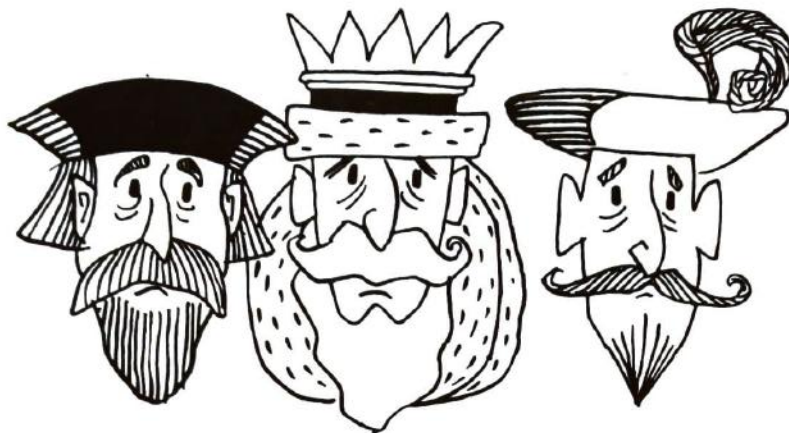
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Revolution in Platonina

After the revolution in Platonina, the dethroned king, his prime minister, and his treasurer were arrested and tried by the revolutionary court. Before sentence was passed, the judge spoke to each of the prisoners.

To the king he said, "If *and only if* both your ministers receive the same sentence will you be executed."

To the prime minister, he said "If *and only if* the king and the treasurer receive the same sentence will you be imprisoned."

To the treasurer, the judge said, "If *and only if* the other two receive different sentences will you be set free."

The judge then pronounced his sentence. "Tomorrow morning one of you will be set free, another will be imprisoned for life, and the third will be executed." He then ordered the guards to lead the prisoners away.

"Wait!" they shouted, "what's to become of us?"

The judge merely replied, "I've already told you," and he walked calmly away. What were the respective sentences?

Solution:

The king was imprisoned: the prime minister was executed: the treasurer was freed. Since all three men received different sentences, the king was not executed and the prime minister was not imprisoned. Thus the king could not have been executed or freed, so he was imprisoned. The prime minister must have received the remaining sentence and was therefore executed.

Puzzles

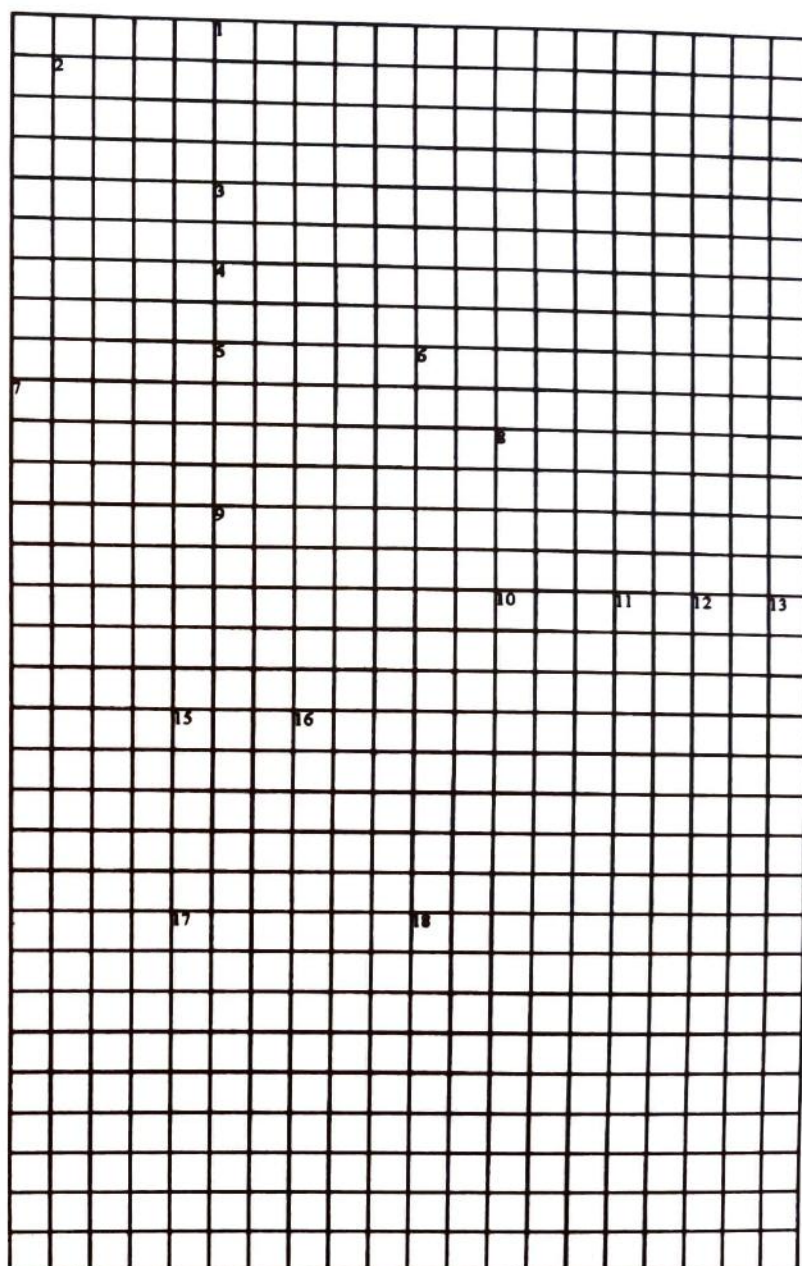
BY TOMMY WU

Across:

2. A line intersecting a circle at one point
3. Inverses of logarithmic functions are _____ functions.
4. Transformation $(x,y) \rightarrow (kx,ky)$
5. Set of all real and imaginary numbers
7. Cipher (i.e. this person is a cipher)
9. Transformation $(x,y) \rightarrow (-y,x)$ is a type of _____.
10. A set of ordered pairs
14. The figure that appears most often in a set of data
17. In a subtraction problem, the number "taken away"

Down:

1. Lines forming right angles are _____
2. Mathematics dealing with right triangles
6. Father of (conventional) geometry
8. Opposite of tautology
11. He developed a formula for the volume of a sphere
12. A transformation that preserves distance
13. Inventor of logarithms
15. A major field of mathematics dealing with the concept of a limit
16. A parabola is equidistant from a point called the focus and a line called the _____.
18. The set of points such that for each point the absolute value of the difference from two given points is a constant



Solutions:



XWTACDEGHGFABQUINTICDCWMLPRQ
 AAZBIYNOITALIDJIAFDEDBGCECA
 AFWGSZBYXCWXDXENCEQUILATERAL
 ICDJOFGZKPMKNLOQORZSYTXUDVW
 NCQUMQYTIHYCACTPMNSESPILLEFG
 YOTHEHGNLHATNTODBSSEOMCFAEC
 ANAQTUOASDMNHWPPIUGOPPQAHRS CB
 XTISRQLDEBODIAQAAJTETIGDGCDC
 ZRDEYAOLAPSDXRGBCEAPRRRIULHTM
 XABELQOTALYGCESSTOFLKQAAANBNIBJ
 WPOCELUTBRNPPTCUARZBAPAAACRBS
 AOYDLIACASXOYRAYGAUCELRDMERM
 CSBABDТАINCITYYHVWSXZZYPDFKH
 BISNPESRAMXIIIOOREDONOUPIRLST
 XTSUDUCFIPAESJNMADVWIVEAOEQI
 PIOQEDZHETEMPSEXORARMDEJSSCLR
 RVIPMPVPNEZAREAAAIRZSOETRYPTPA
 TEBLOYISYLLLOGISMJJYXNIKMTIMG
 OOEURRHKAYUOZNGOFFGFPVWTPQOOO
 QXLMGICLKNOGATPEHSLOEEHTUNNL
 SZUGAEZAI FRIANE G I FLMIGOSHKB
 NATZNBHPARABOLAPEHRTDNSTVOTJ
 URURIRYPERMUTATIONSNUQREXTIR
 WVNZEEOTNRFKXNECITFAATPAPAWPB
 ETANIDROBKPNPXFZDIRECTRIXYAUQ
 MBREDOIECIFUGHDWNYTESPZLIZQB
 XCSDOTXLADEMTFRUSVISONRMJOVB
 GGFACONJUGATEAFIGJKOTCWGAKMQ
 YHKNCIWYEJXGAATHXTVCWTHNZLIO
 LIUIOJHPOTENUSEGMUWADI IEMFH
 ZFNPWXUWSHWICYTIVITISNARTGNP

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Math Survey

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